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# Quota baskets versus tax baskets in fisheries

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**Abstract**

Because managing stocks individually is costly, many fisheries bundle multiple stocks and manage them with a single instrument. I study when such bundling is desirable and whether it should take the form of a shared quota or a shared landing fee, extending the constant-escapement framework of Reed (1979) and Weitzman (2002) to a multispecies fishery under ecological uncertainty.

The fundamental condition for a well-designed basket is equalizing marginal profits across species, a condition that combines biological productivity, harvesting costs, and prices. Under this criterion, I find that the tax basket outperforms the quota basket. However, two qualifications refine this result: (i) both instruments coincide and are first-best when marginal profits are equalized at each species' optimal escapement in a deterministic setting; and (ii) as the number of species in the group increases, the quota's uncertainty loss is diversified away, reducing the average per-species welfare gap. Yet increasing the basket size has a second cost: the mismatch from pooling heterogeneous species grows with the number of species for both instruments. Larger baskets are not always better.

Bundling improves species-specific management only when the monitoring cost savings exceed this mismatch loss.

**JEL Classification:** Q22, Q28, Q58, D81

## 1 Introduction

Allocating property rights through species-specific quotas has been shown to mitigate the common-pool resource problem in fisheries (Costello et al., 2016; FAO, 2025; Hilborn et al.,

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2020; Hoshino et al., 2020; Mangin et al., 2018). Without quotas, competing fishermen have an incentive to harvest more today, depleting the biomass available tomorrow (Clark, 1990; Gordon, 1954). Despite the success of catch shares, typically only wealthy countries can bear the costs of fully implementing them for many species (Mangin et al., 2018). Because most commercial fisheries harvest multiple species jointly (Fulton et al., 2022; Spence et al., 2021), some countries instead pool many stocks and regulate them as one (Kulanujaree et al., 2020) or apply a single economic instrument to the whole group, a quota basket (Sanchirico et al., 2006). Quota baskets have historically been dismissed because they risk overfishing vulnerable species (Bonzon et al., 2013; Clark and Kirkwood, 1986), yet there is evidence that they can be economically and ecologically viable (Collado et al., 2021; Karr et al., 2021; Kritzer et al., 2023; Kulanujaree et al., 2020). Bundling trades a blunter instrument for lower administrative costs. How do we reconcile the overfishing risk with the viability observed in practice?

In this paper, I ask when it is desirable to bundle species under a single instrument and whether that instrument should be a shared quota or a uniform landing fee. To compare the two, I introduce the tax basket, a single landing fee applied to a group of species. To my knowledge, no fishery relies exclusively on landing fees, but the tax basket isolates how each instrument handles ecological uncertainty, because the optimal landing fee does not depend on ecological shocks (Weitzman, 2002). I extend Reed's (1979) constant-escapement model and Weitzman's (2002) information framework to a fishery with $N$ heterogeneous, ecologically independent stocks, one fisherman, one social planner, and one basket. The planner sets the instrument before ecological shocks are realized; the fisherman harvests after observing them. Under either instrument, the fisherman equalizes marginal profits across species, so designing a basket amounts to choosing one common margin for all species. I benchmark both baskets against species-specific landing fees, which implement Reed's escapement for every stock, and decompose the welfare loss into a mismatch loss, from imposing one margin on heterogeneous species, and an uncertainty loss, from letting the margin move with realized shocks. To isolate the role of the instrument, the model abstracts from management costs, gear choice, ecological interactions, strategic behavior, and bycatch.

I obtain three main results. First, the welfare-maximizing basket margin is a weighted average of the species' next-period marginal profits, the fisheries analog of the optimal uniform tax across heterogeneous pollutants. As an intertemporal equimarginal principle, marginal profits combine biological productivity, harvesting costs, and prices. Second, Weitzman's ranking survives bundling: the tax basket yields higher expected welfare than the quota basket, because the tax targets the margins directly, while the quota's shadow value moves with the aggregate shock. Third, basket size has two effects. Independent shocks partly cancel in the aggregate, so the quota basket's disadvantage per species falls at rate $1/N$, although its total disadvantage does not vanish. At the same time, the mismatch loss grows roughly linearly with $N$ for both instruments. Simulations at the typical recruitment variability of exploited fish stocks (Thorson et al., 2014) illustrate this

trade-off: the uncertainty loss is the larger of the two in a two-species basket, but the mismatch loss is already twice as large at five species and roughly thirty times as large at fifty. Larger baskets are not always better: bundling pays only when the monitoring savings exceed these efficiency losses.

The case for bundling rests on three difficulties of species-specific regulation in multispecies fisheries, which stem from joint production, limited selectivity, imperfect information, and fishing gear choice (Branch, 2009; Christensen and Pauly, 2004; Copes, 1986; May and Clark, 1979; Nakamura, 2015; Pauly et al., 1998; Sanchirico et al., 2006). First, species-specific regulation ignores natural clusters (shoals) and species interactions through ecology, regulation, and non-selective fishing gear. As a consequence, it omits the trade-off between sustainability and profit maximization created by choke species: the quota of the most restrictive species leads to premature season closures (Abbott and Wilen, 2009) and blocks the extraction and revenue of non-vulnerable species (Hatcher, 2022; Ulrich et al., 2017; Wo et al., 2022). This was the case for co-occurring groundfish fisheries on the U.S. West Coast. Hilborn et al. (2004) modeled twelve species in this fishery and found that maximizing profits meant overfishing some species, whereas preventing overfishing of the most vulnerable stocks caused a 90% loss in yield. Mixed fisheries have repeatedly failed to achieve the intertemporal profit maximization that characterizes species-specific catch-share management.

Second, Tinbergen's (1952) rule calls for one corrective instrument per market failure, but planners are constrained by how many economic instruments they can credibly design, monitor, enforce, and govern. Fully implementing regulation for even one species is data-intensive (Mangin et al., 2018; Su et al., 2020), costly to administer (Karr et al., 2021), and politically unstable (OECD, 2003). Even when the resources are available, species are hard to identify at port or during the fishing season (Pacific Fishery Management Council, 2024; Squires et al., 1998) or are aggregated in datasets (Su et al., 2020; Sun et al., 2023). Such problems are especially pronounced in Chinese fisheries, which often harvest a broad mix of species with minimal selectivity, leading to indiscriminate removal across trophic levels (X. Jin et al., 2020; Kritzer et al., 2023; Szuwalski et al., 2017). Low selectivity hampers control and monitoring (Sun et al., 2023; Ye et al., 2013), while aggregation in official landings data complicates identifying stock and harvest composition (Cao et al., 2017; Pauly and Froese, 2012). These issues limit the feasibility of traditional species-specific management that relies on precise stock assessments.

Third, applying species-specific instruments to a multispecies group goes beyond administrative difficulty: the mismatch also causes concrete harm. It induces discarding of low-value species, or high-grading (Copes, 1986; Ulrich et al., 2017), effort reallocation toward unregulated stocks (Smith et al., 2009), and intertemporal arbitrage (Holzer and DePiper, 2019).

This aggregation problem belongs to a broader class of second-best regulation problems in which a single instrument must address multiple market failures. In environmental economics, this question has been studied for uniform versus differentiated taxes across

heterogeneous pollution sources in a static setting (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986), for the choice between prices, quantities, and mixed policies when multiple pollutants are regulated jointly (Ambec and Coria, 2013), and for single taxes versus single quotas under uncertainty (Hoel and Karp, 2002). The fisheries literature has also proposed alternatives for handling multiple stocks, such as a shared cap set as a percentage of the sum of individual caps (Sanchirico et al., 2006), a shared cap that targets a group mortality (Garcia et al., 2020), a shared cap with safeguard caps for vulnerable species (Sun et al., 2025), multispecies maximum sustainable yield (Fulton et al., 2022), aggregate production models (Fogarty et al., 2012), quota exchanges (Singh and Weninger, 2024), and size-spectrum models (Andersen et al., 2016), among others. Most of these methods help, but they are prohibitive for many regulators because of limited scientific capacity, information, enforceability, and budget. Moreover, many do not incorporate ecological uncertainty.

Unlike these solutions, quota baskets do not rely on a single-species framework or rich datasets (Collado et al., 2024). They bundle species into a single instrument, allowing flexibility in harvest allocation by trading stringency for convenience. However, theoretical and empirical research on this topic remains scarce. Clark (1990) showed that joint harvesting could not sustain both species in the short run, depleting the more vulnerable stock in a static open-access setting. Extending the analysis dynamically, he demonstrated that joint harvesting may be feasible but leads to an unstable equilibrium. In a non-fishery setting, Weitzman (1998) examined biodiversity preservation under budget constraints and showed that optimal bundling strategies may imply sacrificing some species to preserve others. More recently, Collado et al. (2021) analyzed the fisherman's choice of fishing gear under multiple basket allocations in a modified Gordon–Schaefer model. The authors suggest that well-designed baskets can maintain moderate overall stock survival and intertemporal profit maximization, whereas poorly designed configurations risk the collapse of the vulnerable species or of all species in the group.

Finally, consensus remains lacking on how to group species and on the conditions needed for a welfare-maximizing quota basket. Previous research recommends grouping species by similarity in ecological and economic traits (Collado et al., 2021; Costello and Szuwalski, 2018), by similar vulnerability indicators (catchability over intrinsic growth rate) (Collado et al., 2024), or by relying on local fishermen's expertise (EDF, 2020). Using empirical data from Belizean fisheries (United Nations Conference on Trade and Development, 2022), Collado et al. (2024) found that the optimal quota basket was approximately 30% of the sum of individual maximum sustainable yields for selected baskets. These efforts build intuition but do not provide a definitive answer, especially in more complex settings such as those with ecological uncertainty.

To my knowledge, this is the first paper to compare quota baskets with tax baskets. It extends the seminal work of Reed (1979) and Weitzman (2002) to a multispecies setting and speaks to the literature on regulating multiple pollutants (Ambec and Coria, 2013; Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). The optimal basket margin

is the fisheries analog of the optimal uniform tax across heterogeneous pollutants, where the tax is a weighted average of marginal damages. Most of that literature is static; this paper studies a dynamic, stochastic setting. Diversification across species also links basket design to portfolio theory (Edwards et al., 2004; D. Jin et al., 2016; Markowitz, 1952; Sanchirico et al., 2008), but here adding species reduces variance rather than optimizing portfolio weights.

The paper also contributes to the literature on partial flexibility in multispecies quota systems. Whereas Singh and Weninger (2024) characterize the optimal degree of flexibility given a species-specific quota design, I derive the conditions for full bundling and compare the resulting quota basket with its tax counterpart. The resulting grouping criterion, similarity in individual optimal taxes, offers an alternative to vulnerability indices (Collado et al., 2024) that combine catchability and intrinsic growth.

Finally, the paper contributes to the literature on fisheries under uncertainty (Clark and Kirkwood, 1986; Hannesson and Kennedy, 2003; Reed, 1979; Sethi et al., 2005). The analysis shows that, even under uncertainty, basket instruments can be designed using a well-defined optimality criterion rather than ad hoc grouping rules. However, I provide a condition that narrows the gap between instruments to a constant. When taxes are not politically viable, it is useful to improve the effectiveness of quotas.

The remainder of the paper is organized as follows. Section 2 presents the model and theoretical analysis. Section 3 reports the numerical results. Section 4 discusses policy implications and concludes.

## 2 Model

Throughout the paper, $t = 0, 1, \ldots$ denotes time, and $i = 1, 2, \ldots, N$ indexes the different species. The model relies on two cornerstones: (i) the intertemporal profits for a particular species are maximized with a fixed escapement over time in both deterministic and stochastic settings (Reed, 1979), and (ii) a fixed landing fee induces the fisherman to stop fishing at the optimal escapement (Weitzman, 2002).[1] I model one basket, one fisherman, and one social planner or regulator.

For the ecology component, I make three key assumptions. First, I assume the species' life cycles are independent, so each species' growth function depends only on its own remaining stock (escapement) $y^i_{t-1}$ from the previous period. The function follows the usual conventions: it is differentiable, concave $\frac{\partial^2 G^i(y^i_{t-1})}{\partial (y^i_{t-1})^2} \leq 0$, and increasing $\frac{\partial G^i(y^i_{t-1})}{\partial y^i_{t-1}} > 0$. Second, I assume the random ecological shocks $\{Z^i_t\}$ are independent and identically distributed over time on some interval $[a, b]$,[2] with $\mathbb{E}[Z^i_t] = 1$ and $\mathbb{E}|\ln Z^i_t| < \infty$. Third, I assume the

[1] An alternative proof can be found in Appendix A.

[2] The population is self-sustaining within the interval $[\underline{x}, \overline{x}]$, where $\overline{x} = \max\{bG(s) : bG(s) \geq s, s \geq 0\}$ and $\underline{x} = \max\{aG(s) : aG(s) \geq s, s \geq 0\}$.

shocks are exogenous to the stocks and the choice variables.[3]

For the fisherman's behavior, I make three key assumptions. First, harvesting costs can differ across species within a basket (through the cost coefficients $\theta^i$); I do not model the choice of fishing technology or gear. Second, prices are strictly positive for all species in the basket; therefore, discards are not considered, as all harvest is valuable. Finally, all species can be fished in all periods (the fishable set $\mathcal{F}$ defined in Lemma 1).

For the market component, I assume prices $p^i$, cost parameters $\theta^i$, and the discount factor $\beta$ are time-invariant. I also assume the marginal harvesting cost is positive and decreasing in the available stock $s^i$: $C^i(s^i) > 0$, $C^{i\prime}(s^i) < 0$. I work with the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, which corresponds to constant catchability. I show that this form lets the tax deliver the manager's target escapements regardless of the realized shocks.

I also adopt Weitzman's (2002) information framework, which combines a static optimization problem for a myopic fisherman with a forward-looking social planner:

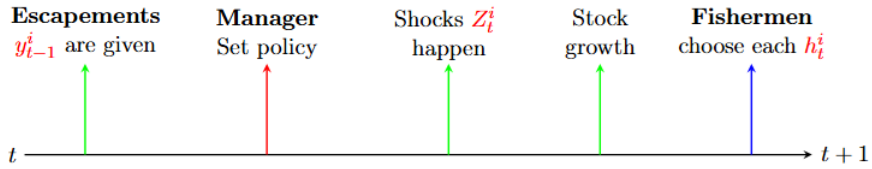



Figure 1: Information flow for each species $i$ between periods $t$ and $t+1$

At each period $t$, the following sequence of events occurs:

1. Escapements $y^i_{t-1}$ for each species $i$ are predetermined by the choices of the previous period.
2. The social planner sets a basket policy (a quota $\omega_t$ or a tax $\tau_t$) without knowing how the stock composition will change with the ecological shocks.
3. The ecological shocks $Z^i_t$ are realized. Each species grows, and the stock shares change.
4. The fisherman observes the realized stocks and chooses the harvest $h^i_t$, given the basket policy set by the planner.

The key friction is that the planner chooses a basket policy to address multiple common-pool resource problems before shocks are realized, whereas the fisherman makes harvest decisions after shocks are realized. Given the economic instrument, a myopic fisherman observes the realized shocks and chooses harvest to maximize current profit:

[3] That is, $\mathbb{E}[Z^i_t \mid \cdot] = 1$ conditional on the present and past values of $X^i_t$ and of the choice variables.

1. Under a tax basket:

$$\max_{\{h_t^i\}_{i=1}^N} \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} (p^i - \tau_t - C^i(s^i))\, ds^i \tag{1}$$

2. Under a quota basket:

$$\begin{aligned} &\max_{\{h_t^i\}_{i=1}^N} \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} (p^i - C^i(s^i))\, ds^i \\ &\text{subject to} \quad \omega_t = \sum_{i=1}^{N} h_t^i \end{aligned} \tag{2}$$

where $X_t^i$ is the stock at the start of period $t$, and $h_t^i$ is the harvest. Because the escapement $y^i$ is the reference for species-specific profit maximization, I re-express the integral bounds using the definition of escapement, $y_t^i = X_t^i - h_t^i$:

$$\sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} \pi^i(s^i)\, ds^i = \sum_{i=1}^{N} \int_{y_t^i}^{X_t^i} \pi^i(s^i)\, ds^i$$

The forward-looking social planner, in turn, sets the instrument ($\omega_t$ or $\tau_t$) before shocks are realized to maximize expected discounted profits, taking the fisherman's decision rule for each species $i$ as given:

1. Under a tax basket:

$$\begin{aligned} &\max_{\{\tau_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \sum_{i=1}^{N} \left( \underbrace{\int_{X_t^i - h_t^i(\tau_t)}^{X_t^i} (p^i - \tau_t - C^i(s^i))\, ds^i}_{\text{Species } i \text{ profit}} + \underbrace{\int_{X_t^i - h_t^i(\tau_t)}^{X_t^i} \tau_t\, ds^i}_{\text{Species } i \text{ tax revenue}} \right)\right] \\ &\text{subject to} \quad X_t^i = Z_t^i G^i(y_{t-1}^i) \end{aligned} \tag{3}$$

   The tax terms in the profit and revenue components cancel.

2. Under a quota basket:

$$\begin{aligned} &\max_{\{\omega_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \sum_{i=1}^{N} \int_{X_t^i - h_t^i(\omega_t)}^{X_t^i} (p^i - C^i(s^i))\, ds^i\right] \\ &\text{subject to} \quad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad \omega_t = \sum_{i=1}^{N} h_t^i \end{aligned} \tag{4}$$

where $\beta \in (0,1)$ is the discount factor.

With this procedure, I obtain (1) the fisherman's static decision rule for $y_t^i$, (2) the planner's dynamic optimal policy choice after incorporating (1), and (3) the fisherman's escapement response to the choice in (2). I can then calculate the welfare difference between instruments. To complete the comparison, I use the species-specific first-best taxes $\tau^{i*}$, which Weitzman (2002) shows do not depend on ecological shocks. $\mathcal{W}^{\text{ind}}$ denotes the expected intertemporal sum of profits and tax revenues in this case. The tax basket applies a uniform landing fee $\tau^+$ to the group and yields welfare $\mathcal{W}^\tau$; the quota basket applies a single aggregate harvest limit $\omega_t$ and yields welfare $\mathcal{W}^\omega$. Because the first-best vector of taxes avoids both the uncertainty loss and the mismatch loss (the efficiency loss from forcing a common instrument level on all species), and because I will show that the tax basket avoids the uncertainty loss, the three schemes are ordered $\mathcal{W}^{\text{ind}} \geq \mathcal{W}^\tau \geq \mathcal{W}^\omega$. Assuming a risk-neutral regulator, I decompose the regulatory loss as

$$\underbrace{\mathcal{W}^{\text{ind}} - \mathcal{W}^\omega}_{\text{Regulatory loss}} = \underbrace{\mathcal{W}^{\text{ind}} - \mathcal{W}^\tau}_{\text{Mismatch loss}} + \underbrace{\mathcal{W}^\tau - \mathcal{W}^\omega}_{\text{Uncertainty loss}} \geq 0. \tag{5}$$

The model assumes nothing about the social planner's knowledge of the stock composition at the start of period $t$. This is deliberate: Section 2.3 shows that the manager does not need data on the initial composition of the biomass.

## 2.1 Fisherman's decision rule

From the FOC of Equation 1 with respect to each harvest, and using the definition of escapement $y_t^i = X_t^i - h_t^i$, I obtain the fisherman's optimality condition for the current period:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = \tau_t \tag{6}$$

Equation 6 is the equimarginal principle: total profits are maximized when the marginal profit of the last harvested unit is equal across species. The fisherman's escapement decision is:

$$\begin{aligned} p^i - \tau_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - \tau_t \\ y_t^i &= (C^i)^{-1}(p^i - \tau_t) \end{aligned} \tag{7}$$

In this static problem, the escapement rule also serves as the harvest rule for each species. Given the parameter assumptions, the tax does not depend on the state variables (I verify this in the planner's problem).

In Equation 2, the fisherman faces the same maximization problem but for a given quota basket $\omega_t$. The optimality condition in this case is:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = \lambda_t \tag{8}$$

This yields the same escapement rule, $y_t^i = (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t)$. Note, however, that the shadow value is a function of the stock vector $\vec{X}_t$ and the quota basket $\omega_t$, because

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\ \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \lambda_t &= \lambda(\vec{X}_t, \omega_t) \end{aligned} \tag{9}$$

The fisherman allocates harvest by equalizing marginal profits across all species in the basket at a common margin $m\pi_t$, which equals $\tau_t$ under the tax and $\lambda_t(\vec{X}_t, \omega_t)$ under the quota. The fisherman's decision under a quota basket is therefore:

$$y_t^i = (C^i)^{-1}(p^i - \lambda(\vec{X}_t, \omega_t)) \tag{10}$$

In this case, the shadow value depends on the state of all stocks, which makes the quota basket more complex than the tax basket. In both cases, the variable that drives the fisherman's decision is the marginal profit, and the tax and the quota are vehicles to set it:

**Lemma 1** (Fisherman's optimality condition)**.** *The optimality condition for harvesting $N$ species in the fishable set $\mathcal{F} = \{i : p^i - m\pi > C^i(K^i)\}$, where $K^i$ is the carrying capacity of species $i$, is:*

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = m\pi_t$$

The stock-dependent cost function incorporates the economic and biological parameters of all species into the optimality condition. The set $\mathcal{F}$ is defined at carrying capacity; whether a species is actually harvested in period $t$ also requires $X_t^i > Y^i(m\pi_t)$. The theory assumes this holds every period. Section 3 reports how often it fails under the quota.

## 2.2 Optimal marginal profits

The social planner incorporates the fisherman's escapement decision into her dynamic optimization problem:

$$\begin{aligned} V(\vec{y}_{t-1}) = \max_{m\pi_t \geq 0} \; & \mathbb{E}\left\{ \sum_{i=1}^{N} \int_{Y^i(m\pi_t)}^{Z_t^i G^i(y_{t-1}^i)} \left(p^i - C^i(s^i)\right) ds^i + \beta V(\vec{y}_t) \right\} \\ \text{s.t. } & X_t^i = Z_t^i G^i(y_{t-1}^i), \quad y_t^i = Y^i(m\pi_t) \end{aligned} \tag{11}$$

Taking the FOC with respect to $m\pi_t$ gives the optimal marginal profit:

**Proposition 1** (Welfare-maximizing $m\pi$)**.** *Consider $N$ heterogeneous, independent stocks with cost functions $C^i$, i.i.d. ecological shocks $Z_t^i$ with $\mathbb{E}[Z_t^i] = 1$, and growth functions $G^i(y_{t-1}^i)$. Assume the interior regularity condition*

$$\max_i \tau^{i*} \;<\; \tau^F_{\max} \;:=\; \min_i\Big(p^i - C^i(K^i)\Big), \quad K^i \text{ is the carrying capacity,} \tag{12}$$

*so that all species remain in the fishable set on $[\min_i \tau^{i*}, \max_i \tau^{i*}]$. Then the optimal $m\pi_t^+$ solves the implicit equation*

$$m\pi_t^+ = \mathbb{E}\Big[\sum_{i=1}^{N} \underbrace{\rho_t^i(m\pi_t^+)}_{\text{Weight } i} \quad \underbrace{\pi_{t+1}^i}_{\text{Marginal profit } i \text{ at } t+1}\Big] \tag{13}$$

*where $\rho_t^i = \frac{y^i_{m\pi_t^+}}{\sum_{j=1}^N y^j_{m\pi_t^+}}$, with $y^i_{m\pi} := dY^i/dm\pi$ the response of species $i$'s escapement to the margin, and $\pi_{t+1}^i = \beta\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)$ is the discounted marginal value of one more unit of escapement of species $i$.*

The implicit equation above results from differentiating the Bellman equation and using the fisherman's condition $y_t^i = (C^i)^{-1}(p^i - m\pi_t)$ (see (38) in Appendix B.2). This Euler equation balances the current marginal profit with the weighted average of next-period marginal profits.

**Corollary 1.1** (The optimal $m\pi_t$ is constant)**.** *The optimal marginal profit is constant over time.*

*Proof.* Because the shocks are i.i.d. and $Z_{t+1}^i$ is independent of $y_t^i$, the expectation $\mathbb{E}[\pi_{t+1}^i]$ is a deterministic function of $y_t^i$ alone:

$$\pi_{t+1}^i(y_t^i) := \beta\,\mathbb{E}\Big[\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i\Big] G^{i\prime}(y_t^i).$$

Substituting $y_t^i = Y^i(m\pi_t)$, Equation (13) becomes

$$m\pi_t = \sum_{i=1}^{N} \rho^i(m\pi_t)\, \pi_{t+1}^i\big(Y^i(m\pi_t)\big),$$

an equation in $m\pi_t$ alone that contains no state variable and no time index. Its solution (unique under Corollary 2.1) is therefore the same in every period, and $m\pi_t$ does not depend on $\vec{X}_t$. □

From this result, I obtain the following remarks:

**Remark 1.1** (Deterministic case)**.** *If $Z_t^i \equiv 1$, then $m\pi_t = m\pi_{t+1} = m\pi$.*

*Proof.* With $Z_t^i \equiv 1$, the expectation in (13) is degenerate and the argument of Corollary 1.1 applies directly. □

Under a specific stock-dependent cost function, the optimal margin is unaffected by the ecological shocks.

**Corollary 1.2** (Specific cost function recovers the deterministic case)**.** *If the cost function is $C^i(s^i) = \frac{\theta^i}{s^i}$, the optimal margin under uncertainty equals the optimal margin under certainty.*

*Proof.* With $C^i(s) = \theta^i/s$ and $\mathbb{E}[Z_{t+1}^i] = 1$,

$$\mathbb{E}\Big[\Big(p^i - \frac{\theta^i}{Z_{t+1}^i G^i(y_t^i)}\Big) Z_{t+1}^i\Big] G^{i\prime}(y_t^i) = \Big(p^i - \frac{\theta^i}{G^i(y_t^i)}\Big) G^{i\prime}(y_t^i) = \big(p^i - C^i(G^i(y_t^i))\big) G^{i\prime}(y_t^i),$$

so $\pi_{t+1}^i$ in the proof of Corollary 1.1 is the deterministic marginal value, and (13) coincides with its deterministic counterpart. With $C^i(s) = \theta^i/s^\alpha$ and $0 < \alpha < 1$, the shock no longer cancels and the deterministic margin is only a lower bound on the optimal one (see Appendix A). □

**Remark 1.2** (Re-expressing the weights $\rho_t^i$)**.** *For a cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, the weights $\rho_t^i$ can be re-expressed as a function of the escapements and cost parameters: $\rho^i = \frac{(y^i)^2/\theta^i}{\sum_j (y^j)^2/\theta^j}$.*

*Proof.* The re-expression follows from using the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$ and the fisherman's optimality condition $p^i - m\pi_t = C^i(y_t^i)$ in (13) (see expressions (39) and (40)). □

By inspection, the tax basket equals $m\pi_t^+$, which does not depend on the state variables, while the quota depends on the states through a stochastic shadow value. Consequently, the fisherman's escapement is deterministic under a tax ($\vec{y}$) and stochastic under a quota ($\vec{y}_t$). In a deterministic setting, however, the shadow value is also deterministic, because $\lambda(\vec{X}_t, \omega_t) = \lambda(\vec{G}(\vec{y}_{t-1}), \omega_t)$. The shadow value and the tax basket then behave the same way: $\tau_t = \lambda(\omega_t)$. In addition, if all species have the same marginal profit at their respective optimal escapements, the basket instruments are first-best:

**Corollary 1.3** (First-best basket instruments)**.** *Given the core assumptions, the tax basket implements the first-best if and only if $p^i - C^i(y^{i*}) = p^j - C^j(y^{j*})$ for all $i \neq j$, that is, if all species-specific optimal taxes coincide. In a deterministic setting, the same condition makes the quota basket first-best.*

*Proof.* If species have identical marginal profits at their respective optimal escapements, Reed's optimality condition, which under $C^i = \theta^i/s$ and $\mathbb{E}[Z^i] = 1$ reads $\beta\big(p^i - C^i(G^i(y^{i*}))\big) G^{i\prime}(y^{i*}) = p^i - C^i(y^{i*})$, holds for every species at the common margin:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = \tau_t = \lambda(\omega_t)$$
$$p^1 - C^1\left(y^{1*}\right) = p^2 - C^2\left(y^{2*}\right) = \cdots = p^N - C^N\left(y^{N*}\right) = \tau^* = \lambda(\omega_t)$$
$$\beta\big(p^1 - C^1(G^1(y^{1*}))\big)G^{1\prime}(y^{1*}) = \cdots = \beta\big(p^N - C^N(G^N(y^{N*}))\big)G^{N\prime}(y^{N*}) = \tau^* = \lambda(\omega_t)$$

Even if species' profits are heterogeneous, a fixed tax or shadow value then achieves Reed's constant escapement for every species. Because the tax does not depend on the shocks, the result for the tax basket also holds under uncertainty. □

This analysis is incomplete: I still need to prove formally that $\tau^+ = m\pi_t^+$, and to show whether choosing the optimal shadow value is equivalent to choosing the optimal quota $\omega_t^+$ that delivers $m\pi^+$. In the next section, I prove them.

### 2.3 Planner's problem

After studying the fisherman's decision and recognizing how the dynamic choice of $m\pi^+$ shapes the harvest decision, the planner chooses the level of the tax or quota basket to maximize intertemporal profits. For the tax basket, the solution of the dynamic problem is straightforward because it follows the same procedure as in Appendix B after replacing $m\pi_t = \tau_t$. Because the tax is independent of the state variables in both the fisherman's and the planner's problems, it satisfies the same optimality condition as the welfare-maximizing marginal profit, Equation (38) in Proposition 1.

**Proposition 2** (Welfare-maximizing tax basket)**.** *Given the assumptions for $N$ heterogeneous and independent stocks, species-specific optimal taxes $\tau^{i*}$, and the interior regularity condition*

$$\max_i \tau^{i*} \;<\; \tau^F_{\max} \;:=\; \min_i\big(p^i - C^i(K^i)\big),$$

*which keeps all species in the fishable set on $[\min_i \tau^{i*}, \max_i \tau^{i*}]$, the welfare-maximizing tax basket $\tau^+$ is characterized by the same implicit equation as in Proposition 1:*

$$\tau_t^+ = \mathbb{E}\Big[\sum_{i=1}^{N} \underbrace{\rho_t^i(\tau_t^+)}_{\text{Weight } i} \quad \underbrace{\pi_{t+1}^i}_{\text{Marginal profit } i \text{ at } t+1}\Big] \tag{14}$$

*where $\rho_t^i = \frac{y^i_{\tau_t^+}}{\sum_{j=1}^N y^j_{\tau_t^+}}$ is the per-species weight, and $\pi_{t+1}^i(\tau^+) = \beta\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big)Z_{t+1}^i G^{i\prime}(y_t^i)$ is the next-period marginal profit.*

*Welfare is strictly lower than the summation of species-specific first-best whenever the $\tau^{i*}$ are not all equal. Then, $\tau^+$ is the second-best. The tax basket $\tau^+$ also satisfies Corollaries 1.1 to 1.3.*

*Proof.* Replacing $m\pi_t^+ = \tau_t^+$ in Proposition 1 and its corollaries shows that the optimal tax basket satisfies the same Euler equation as the optimal marginal profit and is fixed over time (for a complete derivation, see Appendix B.3). □

Decomposing the per-period profits,

$$\int_{y_t^i}^{X_t^i} \left(p^i - \frac{\theta^i}{s}\right) ds \;=\; p^i X_t^i - \theta^i \ln X_t^i - p^i y_t^i + \theta^i \ln y_t^i, \tag{15}$$

substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, and using $\mathbb{E}[Z_t^i] = 1$ and $\mathbb{E}|\ln Z_t^i| < \infty$, I recover Proposition 2: the same problem repeats each period, and it has the same solution even in a stochastic setting (more details in Appendix C). If the condition of Corollary 1.3 holds, the tax basket is first-best because all individual optimal taxes are equal. Otherwise, the tax basket is a second-best policy because it implements, for every species, a constant escapement that differs from its individual Reed escapement. It maximizes profits with a single instrument, but it does not achieve the outcome of species-specific management.

**Corollary 2.1** (Existence, uniqueness and interiority of $\tau^+$)**.** *Under Proposition 2, the regularity condition* $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$*, the cost function* $C^i = \theta^i/s^i$*, and the second-order condition*

$$y\,\pi_{t+1}^{i\prime}(y) + 2\pi_{t+1}^i(y) \;\le\; p^i + \tau, \tag{16}$$

*the optimal tax basket exists, is unique, and is interior.*

The full proof is in Appendix D. Equation (16) (the same as (46)) is a second-order condition that guarantees global concavity and complements the fishable-set assumption to deliver a unique optimal tax basket. The fishable-set condition keeps every species harvested as $\tau$ varies, and (16) keeps each species' contribution concave. Together, they guarantee that a basket is not too heterogeneous, which formalizes the similarity condition for species within a bundle (Collado et al., 2021, 2024; Costello and Szuwalski, 2018; Kritzer et al., 2023).

Taken together, these results show that tax baskets preserve the constant-escapement structure under uncertainty but generally implement a second-best allocation when species differ in marginal profits. First-best outcomes arise only when marginal profits are identical across species at their respective optima. For the policymaker, data on the economic and ecological parameters are enough to determine the optimal tax basket. However, the planner needs current stock data if, in some period, a species' realized stock falls below its target escapement, $X_t^i \le Y^i(\tau^+)$, so that the species drops out of the harvest. In that case, the interiority condition breaks down.

In contrast, the quota basket requires more information because it changes over time. The essential difference from the tax basket is that a single quota constrains only aggregate harvest, so the split across species is the fisherman's choice. Thus, the shadow value $\lambda_t$ is

not a policy instrument but an equilibrium object that depends on the current stocks and the quota, $\lambda_t = \lambda(\mathbf{X}_t, \omega_t)$.

**Proposition 3** (Welfare-maximizing quota basket)**.** *Under the setup of Proposition 2, the welfare-maximizing quota $\omega^+$ satisfies*

$$\mathbb{E}\Big[\underbrace{\lambda(\mathbf{X}_t, \omega_t)}_{\text{Current shadow value}} - \beta \sum_i \rho_t^i \underbrace{V_{y^i}(\vec{y}_t)}_{\text{Continuation value}}\Big] = 0, \tag{17}$$

*where $\lambda_t$ is the fisherman's shadow value, $V$ is the continuation value function, and $\rho_t^i = \frac{Y_\lambda^i(\lambda_t)}{\sum_j Y_\lambda^j(\lambda_t)}$ is the per-species reallocation weight.*

The derivation is in Appendix E.3. Expression (17) equates, in expectation, the current shadow value of an extra unit of aggregate harvest with the $\rho$-weighted continuation value of the escapement. By the envelope theorem, the marginal value of escapement of species $k$ is the value of one extra fish in stock $k$, minus the shadow value $\lambda_t$ of the escapement it displaces, plus its continuation value:

$$V_{y_{t-1}^k}(\vec{y}_{t-1}) = \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \tag{18}$$

Separating the optimality condition into two parts is useful because (17) is the rule for the level of the quota basket, and (18) reveals what happens when the shadow value targeted by the manager, $\lambda_t^+$, differs from the effective shadow value the fisherman selects after the shocks, $\lambda_t^F$. Before discussing how the planner's inability to observe the ecological shocks causes a mismatch between these shadow values, I establish that choosing $\lambda^+$ is equivalent to choosing $\tau^+$:

**Corollary 3.1** (Duality of $\lambda^+$ and $\tau^+$)**.** *Choosing the optimal quota $\omega_t^+$ is equivalent to choosing an optimal shadow value $\lambda^+$, in the sense that the planner's first-order condition reduces to*

$$\lambda^+ = \beta \mathbb{E}\Big[\sum_{i=1}^N \big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i) \frac{y_{\lambda^+}^i}{\sum_{j=1}^N y_{\lambda^+}^j}\Big], \tag{19}$$

*which is identical in form to the tax condition (14). Both instruments target the same escapement vector $\bar{y}^i = \theta^i/(p^i - \tau^+)$.*

The equivalence is demonstrated in Appendix E. Thanks to duality, it is straightforward to establish that (i) $\lambda^+$ is fixed over time; (ii) $\lambda^+$ exists, is unique, and is interior under the same conditions as $\tau^+$; and (iii) the quota $\omega_t$ is not constant. In other words, the two instruments coincide at the planner's optimum but differ in delivery. A regulator who

targets $\lambda^+$ sets the quota basket before the shocks, as the difference between the expected aggregate stock and the sum of target escapements:

$$\omega_t^+ = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\lambda^+), \qquad \sum_i G^i(y_{t-1}^i) = \mathbb{E}_{t-1}\Big[\sum_i X_t^i\Big]. \tag{20}$$

If the planner could observe the realized stocks $X_t^i$ instead, the quota $\sum_i X_t^i - \sum_i Y^i(\lambda^+)$ would induce $\lambda_t^F = \lambda^+$ in every state, and the two instruments would coincide. I refer to (20) as the certainty-equivalent quota.

As with a tax basket, determining the level of a quota basket does not require knowing the stock composition across species, but it does require knowing total biomass. By the end of period $t$, comparing a quota basket under complete information with one under asymmetric information shows the difference between the fisherman's shadow value $\lambda_t^F$ and the planner's $\lambda^+$:

$$\sum_i (Z_t^i - 1)\, G^i(y_{t-1}^i) = \sum_i \Big[(C^i)^{-1}(p^i - \lambda_t^F) - (C^i)^{-1}(p^i - \lambda^+)\Big], \tag{21}$$

so $\lambda_t^F$ inherits the randomness of the aggregate realized stock. The following corollary identifies the source of the distortion:

**Corollary 3.2** (The distortion between $\tau^+$ and $\omega_t$)**.** *In a stochastic setting, the marginal value of escapement under the quota basket decomposes as*

$$\beta V_{y^k}(\vec{y}_{t-1}) = \pi_{t+1}^k(y_{t-1}^k) - \beta\,\mathrm{Cov}\big(R_t, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k), \qquad R_t := \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t), \tag{22}$$

*where the distortion is the covariance between the residual of the planner's FOC, $R_t$, and the ecological shock of species $k$.*

The full proof is available at Appendix E.5. The covariance vanishes only in the deterministic case, where $\lambda_t^F = \lambda^+$ every period. When the species-specific optimal taxes coincide, the tax basket is first-best (Corollary 1.3) but the quota basket is not, because $\lambda_t^F$ still moves with the realized shocks. The next subsection shows that, under uncertainty, the tax basket weakly outperforms the quota basket.

## 2.4 Welfare comparison

To compare the welfare outcomes across instruments, I borrow equation 41 (see the details in Appendix C.1)

$$\mathcal{W} = \sum_i \underbrace{[A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \underbrace{[\phi^i(y_t^i) + \beta a^i(y_t^i)]}_{=:w^i(y_t^i)}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$ and $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$. Because $\Xi_0$ and the shock path are common to both instruments, I can cancel them in the difference. Then, comparing the per-species payoffs $w^i$:

$$w^{i\prime}(y) = \pi^i_{t+1}(y^i) - \big(p^i - C^i(y^i)\big), \qquad w^{i\prime\prime}(y) = \phi^{i\prime\prime}(y) + \beta a^{i\prime\prime}(y) < 0 \tag{23}$$

So $w^{i\prime}(y^i) = 0$ when $\pi^i(\bar{y}^i) = p^i - C^i(\bar{y}^i) = \tau^+$, generating the vector of escapements $\bar{y}$ that maximizes the intertemporal welfare. Under the tax, $y_t^i = Y^i(\tau^+) = \bar{y}^i$ is a fixed number, and the objective separates into a deterministic sequence of static problems for each $i$:

$$E[\mathcal{W}^\tau] = \Xi_0 + \frac{1}{1-\beta} \sum_i w^i(\bar{y}^i)$$

The quota does not separate. The escapement is random $y_t^i = Y^i(\lambda_t^F)$, function of every species' stock, with $\lambda_t^F = \lambda(\mathbf{X}_t, \omega_t^M)$ solving the extraction after the realization of the shocks $Z_t^i$. The vector of random escapements $y_t$ is located in a random hyperplane $\mathcal{H}$. In contrast, the deterministic hyperplane $\mathcal{H}^* := \mathcal{H}_t\big|_{Z_t^i=1}$ corresponds to the tax basket's $\bar{y}$. Through duality, the planner sets $\omega_t$ to target $\mathcal{H}^*$, but the goal is not always reached:

**Proposition 4** (The welfare is higher under a tax basket)**.** *Let both instruments implement the same optimal shadow value $\lambda^+ = \tau^+$ (Corollary 3.1). Under the interior regularity condition and i.i.d. shocks with $E[Z_t^i] = 1$,*

$$E[\mathcal{W}^\tau] \;\geq\; E[\mathcal{W}^\omega]$$

*Proof.* The tax delivers $\sum_i w^i(\bar{y}^i)$ while the quota delivers $E\big[\sum_i w^i(Y^i(\lambda_t^F))\big]$. Since $\Xi_0$ cancels, summing the discounted per-period gaps,

$$E[\mathcal{W}^\tau] - E[\mathcal{W}^\omega] = \sum_{t\geq 0} \beta^t \, E\Big[\sum_i \big(w^i(\bar{y}^i) - w^i(y_t^i)\big)\Big]$$

Because $w^i$ is strictly concave with global maximizer $\bar{y}^i$ (23), Jensen's inequality gives

$$E\Big[\sum_i w^i\big(Y^i(\lambda_t^F)\big)\Big] \;\leq\; \sum_i w^i\big(E[Y^i(\lambda_t^F)]\big) \;\leq\; \sum_i w^i(\bar{y}^i),$$

Equivalently, by Appendix E.5, the tax attains $\beta V^\tau_{y^k} = \pi^k(y^k_{t-1})$ exactly while the quota attains $\beta V^\omega_{y^k} = \pi^k(y^k_{t-1}) - \beta \operatorname{Cov}(R_t, Z_t^k) G^{k\prime}$. Because $\lambda_t^F$ is a continuous random variable, $\Pr(\lambda_t^F = \lambda^+) = 0$, so $\Pr(y_t^i = \bar{y}^i) = 0$. Hence $\mathcal{H}_t \neq \mathcal{H}^*$ almost surely. □

Under the assumptions of this paper, Weitzman's (2002) result holds: a tax outperforms a quota even under bundled management. This outcome rests on the cost function because the tax is stock-independent. Appendix B.3 shows that, under a cost function with decreasing catchability, the tax loses robustness and, in consequence, the ranking across

instruments narrows. Also, the per-species welfare gap between instruments diminishes as more species are included:

**Proposition 5** (Diversification of uncertainty loss as $N \to \infty$)**.** *The per-species welfare difference between instruments vanishes as* $N \to \infty$.

*Proof.* Under a quota basket, compare the escapement budget between the stochastic and deterministic settings: $\sum_i y_t^i = \sum_i Z_t^i G^i(y_{t-1}^i) - \omega_t$ and $\sum_i \bar{y}^i = \sum_i G^i(\bar{y}^i) - \omega_t$,

$$\sum_i (y_t^i - \bar{y}^i) = \sum_i [Z_t^i G^i(y_{t-1}^i) - G^i(\bar{y}^i)] = S_t, \qquad S_t \approx \sum_i (Z_t^i - 1) G^i(\bar{y}^i)$$

[4] if the scheme sits at the target. By variance additivity and independence,

$$\mathrm{Var}(S_t) = \sigma_Z^2 \sum_i [G^i(\bar{y}^i)]^2 = \Theta(N),$$

so the species-specific variance becomes $\mathrm{Var}(S_t/N) = \Theta(1/N) \to 0$ by the law of large numbers. Now, note that the escapements depend on the shadow values:

$$\sum_i \left[ Y^i(\lambda_t^F) - Y^i(\lambda^+) \right] = S_t \tag{24}$$

Every $Y^i$ is strictly increasing, so all $N$ differences share the sign of $\lambda_t^F - \lambda^+$ and reinforce rather than cancel. Suppose $\lambda_t^F > \lambda^+$ (the reverse is symmetric). Then, by the mean value theorem, each difference is $Y^{i\prime}(\zeta^i)(\lambda_t^F - \lambda^+) \geq \underline{m}(\lambda_t^F - \lambda^+) > 0$ with the slope bound floor $\underline{m}$ of Lemma 3 (see Appendix F). After summing $N$ deviations, I obtain:

$$S_t \geq \underline{m}\, N\, (\lambda_t^F - \lambda^+) \quad \Longrightarrow \quad |\lambda_t^F - \lambda^+| \leq \frac{|S_t|}{\underline{m}\, N}$$

Squaring and taking expectations,

$$E[(\lambda_t^F - \lambda^+)^2] \leq \frac{\mathrm{Var}(S_t)}{\underline{m}^2 N^2} = \frac{\Theta(N)}{\underline{m}^2 N^2} = \Theta(1/N) \to 0$$

Then, the shadow values converge, $E[(\lambda_t^F - \lambda^+)^2] = \Theta(1/N)$. Since $y_t^i = Y^i(\lambda_t^F)$ and $\bar{y}^i = Y^i(\lambda^+)$, the upper slope bound of Lemma 3 gives:

$$|y_t^i - \bar{y}^i| \leq \overline{m}^i\, |\lambda_t^F - \lambda^+| \quad \Longrightarrow \quad E[(y_t^i - \bar{y}^i)^2] \leq (\overline{m}^i)^2\, E[(\lambda_t^F - \lambda^+)^2] = O(1/N)$$

Because $\bar{y}^i$ is the basket maximum of $w^i$, Lemma 2 bounds the per-species loss by:

$$0 \leq w^i(\bar{y}^i) - w^i(y_t^i) \leq \tilde{B}^i\, (y_t^i - \bar{y}^i)^2, \qquad \tilde{B}^i := \tfrac{1}{2} \max_{[\underline{y}^i, \bar{y}^i]} \left( -w^{i\prime\prime} \right) < \infty$$

[4] For simplicity, I evaluate at near target escapement

Taking expectations,

$$0 \le w^i(\bar{y}^i) - E[w^i(y_t^i)] \le \tilde{B}^i \, E[(y_t^i - \bar{y}^i)^2] = O(1/N)$$

The per-species loss is $O(1/N)$. Summing the $N$ per-species losses,

$$0 \le E[\mathcal{W}^\tau] - E[\mathcal{W}^\omega] = \sum_i \big(w^i(\bar{y}^i) - E[w^i(y_t^i)]\big) \le \sum_i \tilde{B}^i \, E[(y_t^i - \bar{y}^i)^2] = \sum_i O(1/N) = O(1)$$

Adding species diversifies each shock at rate $1/N$, but the total welfare loss persists.

□

Adding more species brings the quota basket's per-species outcomes closer to the tax basket. However, the gap does not disappear as $N \to \infty$. With the comparison between basket instruments, I can now incorporate the first-best outcome:

**Proposition 6** (Comparison between the first-best and basket instruments)**.** *The welfare of managing the species individually (a species-specific first-best tax $\tau^{i*}$ for each stock) is at least as large as the welfare of managing the same species under a single tax basket $\tau^+$, and strictly larger unless all species share the same optimal tax.*

*Proof.* From the welfare decomposition (41), both regimes share the policy-independent constant $\Xi_0 = \sum_i [A^i(X_0^i) - \kappa^i \beta/(1-\beta)]$, and each reduces to a discounted stream of the dynamic per-species payoff $w^i(y) = \phi^i(y) + \beta a^i(y)$. Since escapement is constant over time under a fixed tax, the streams collapse to a $\frac{1}{1-\beta}$ factor, and the difference is

$$\mathcal{W}^{\text{ind}} - \mathcal{W}^\tau = \frac{1}{1-\beta} \sum_{i=1}^{N} \Big[ \, w^i\big(Y^i(\tau^{i*})\big) - w^i\big(Y^i(\tau^+)\big) \, \Big], \tag{25}$$

where $\Xi_0$ has cancelled. By (23), $w^{i\prime}(y) = \pi_{t+1}^i(y) - (p^i - C^i(y))$, which vanishes exactly at the individual optimum: $Y^i(\tau^{i*})$ satisfies Reed's condition. Since $w^i$ is strictly concave, $Y^i(\tau^{i*})$ is its unique global maximizer, so every bracket in (25) is nonnegative:

$$w^i\big(Y^i(\tau^{i*})\big) \ge w^i\big(Y^i(\tau^+)\big) \qquad \text{for all } i,$$

strictly whenever $\tau^+ \neq \tau^{i*}$. Summing and dividing by $1-\beta > 0$ gives $\mathcal{W}^{\text{ind}} - \mathcal{W}^\tau \ge 0$, with equality only when $\tau^{i*} = \tau^+$ for every $i$. Combined with Proposition 4 ($\mathcal{W}^\tau \ge \mathcal{W}^\omega$), this orders the three regimes:

$$\mathcal{W}^{\text{ind}} \;\ge\; \mathcal{W}^\tau \;\ge\; \mathcal{W}^\omega$$

□

The individual tax places each species at the peak of its own dynamic payoff $w^i$. The basket forces a single $\tau^+$ on every species, pushing each off-target stock down its own concave hill. This shortfall is the mismatch loss in (5).

# 3 Numerical results

To verify the welfare comparison, I simulate welfare under each instrument using the same randomly generated fisheries and the same Monte Carlo shocks. I use simulated rather than empirical fisheries so that every basket satisfies the regularity condition by construction.

## 3.1 The dataset

Each replication draws a basket of $N$ species that satisfies the regularity condition (12) under the cost function $C^i = \theta^i / s^i$. Drawing $(r^i, K^i, p^i, \theta^i)$ naively becomes impractical as $N$ grows because the rejection rate explodes, which illustrates that species in a basket cannot be too heterogeneous (Corollary 2.1). To avoid this, I fix each species' Reed escapement as a fraction of its carrying capacity, $\iota^i := y^{i*}/K^i$, and invert Reed's condition for the cost coefficient:

$$\theta^i \;=\; \frac{p^i\left(\frac{1}{\beta} - G^{i\prime}(y^{i*})\right)}{\dfrac{1}{\beta y^{i*}} - \dfrac{G^{i\prime}(y^{i*})}{G^i(y^{i*})}}, \qquad y^{i*} = \iota^i K^i. \tag{26}$$

Holding $p^i$ and $\iota^i$ fixed, the optimal tax does not vary with $K^i$. I therefore draw the parameters first and then compute the cost coefficient $\theta^i$. The ranges are narrow enough that no single species dominates the aggregate. Growth rates and carrying capacities capture biological heterogeneity, and prices capture economic heterogeneity.

Growth shocks $Z_t^i$ are i.i.d. lognormal with unit mean and scale each stock's growth increment, $X_t^i = y_{t-1}^i + Z_t^i\, r^i y_{t-1}^i (1 - y_{t-1}^i/K^i)$. I set the baseline volatility to $\sigma_Z = 0.74$, where $\sigma_Z$ is the standard deviation of $\ln Z_t^i$. This value is the mean standard deviation of log recruitment deviations estimated across exploited marine fish stocks by Thorson et al. (2014).

## 3.2 Monte Carlo simulation

For each $N \in \{2, 5, 10, 20, 30, 50\}$, I draw one basket and simulate all instruments over $T = 40$ periods on 2,000 shock paths (1,000 antithetic pairs). I implement the quota as a cap: the fisherman may land less than $\omega_t$, in which case $\lambda_t^F = 0$.

I use two variance-reduction techniques: (i) common random numbers, so that the tax and the quota face identical shocks; and (ii) antithetic variates, which pair each path drawn with $\varepsilon$ with a path drawn with $-\varepsilon$. Together they reduce the standard error of the estimated welfare gap. I verified that the tax and the quota yield identical welfare in a deterministic setting for heterogeneous baskets. Because shocks affect each instrument differently, I also report a sensitivity analysis for $\sigma_Z \in \{0.2, 0.4, 0.6, 0.74, 1.0\}$.

I consider two basket scenarios: (1) a symmetric basket, in which all species have identical parameters but independent shocks; and (2) a balanced basket, in which species

differ in parameters but no single species dominates (defined formally in Section 3.3.3). Welfare is measured in monetary units, as the expected present value over the $T = 40$ periods.

To decompose regulatory loss, I use two Monte Carlo layers: an inner layer that averages over ecological shock paths while holding a basket fixed, and an outer layer that averages over random baskets. This exercise averages 25 baskets, each simulated on 150 antithetic pairs (300 paths). I need the outer layer because the decomposition compares basket and species-specific instruments, and the latter vary considerably across baskets. Table 1 summarizes the parameters and simulation settings.

| Parameter | Meaning | Range / value |
|---|---|---|
| $r^i$ | intrinsic growth rate (per period) | $\mathcal{U}[0.30, 0.40]$ |
| $K^i$ | carrying capacity (biomass units) | $\mathcal{U}[1200, 2400]$ |
| $p^i$ | price (money per biomass unit) | $\mathcal{U}[95, 105]$ |
| $\iota^i = y^{i*}/K^i$ | escapement fraction | $\mathcal{U}[0.48, 0.52]$ |
| $\theta^i$ | cost coefficient | from (26) |
| $\beta$ | discount factor | $1/1.1 \approx 0.909$ |
| $\sigma_Z$ | s.d. of $\ln Z_t^i$ | $\{0.2, 0.4, 0.6, \mathbf{0.74}, 1.0\}$ |
| $T$ | horizon (periods) | 40 |
| $N$ | basket size | $\{2, 5, 10, 20, 30, 50\}$ |
| Paths (convergence) | paths per basket | 2,000 (1,000 pairs) |
| Outer layer (decomposition) | baskets per $N$ | 25 |
| Inner layer (decomposition) | paths per basket | 300 (150 pairs) |

Table 1: Parameters of the simulated fisheries. Initial stocks are set to the target escapement, $X_0^i = \bar{y}^i$, so both instruments start from the same state. The baseline volatility is in bold.

The fishable-set condition holds for the tax basket in every simulated period. Under the quota basket, ex-post shocks occasionally leave a stock below the escapement implied by the realized shadow value, $X_t^i < Y^i(\lambda_t^F)$. The fisherman then lands none of that species in that period. This corner reflects the quota's response to shocks rather than a change in the basket's composition. At baseline volatility, it occurs in 7% of species-periods at $N = 2$ and 1.4% at $N = 50$ in the balanced basket, and in 8–14% in the dominant-species scenario of Section 3.3.3. Corners become rarer as $N$ grows beyond five because $\lambda_t^F$ becomes more stable.

### 3.3 Results

Panel (a) of Figure 2 shows that the expected welfare difference $\mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega]$ is positive at every $N$ and settles at a positive constant. In the symmetric basket, it is 4,054 monetary units at $N = 2$ and 3,567 at $N = 50$. In the balanced basket, it lies between about 3,300

and 3,650 for $N \geq 10$. Every estimate lies between 49 and 73 standard errors from zero. In relative terms, the quota's loss is 1.9% of the tax basket's welfare at $N = 2$, 0.35% at $N = 10$, and 0.07% at $N = 50$. The quota never outperforms the tax basket, and increasing $N$ drives the total gap to a positive, stable constant.

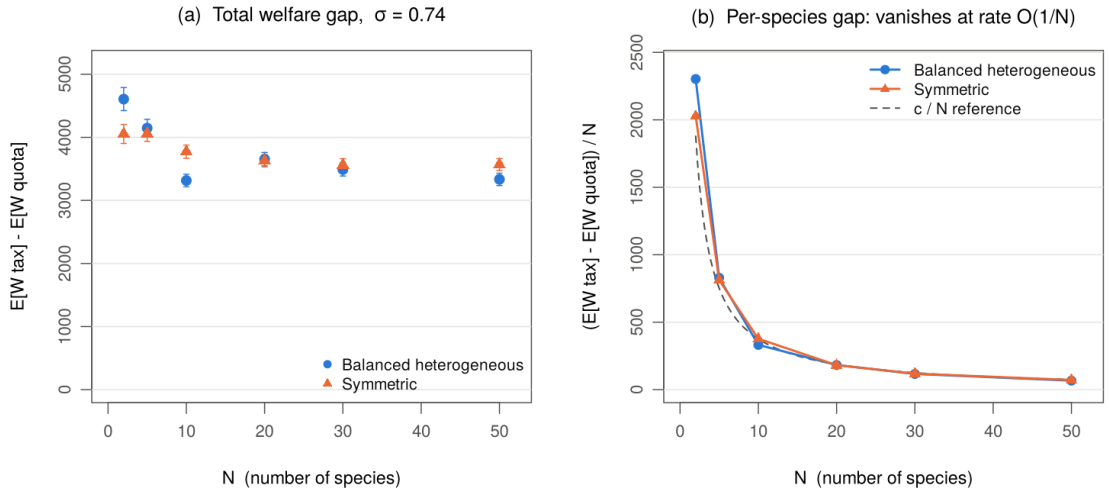



Figure 2: Expected welfare difference $\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}]$ at $\sigma_Z = 0.74$. One basket per $N$, 2,000 shock paths. The bars are $\pm 1.96$ standard errors. Panel (a): the total difference converges to a positive constant. Panel (b): the per-species difference vanishes at rate $O(1/N)$.

Panel (b) divides the result in panel (a) by $N$: the per-species difference vanishes at rate $O(1/N)$, following the $c/N$ reference curve. This pattern is consistent with the quota's realized escapements converging to the tax basket's targets $\vec{y}$ as $N$ grows (Proposition 5). Increasing $N$ reduces the per-species gap in both the symmetric and the balanced basket.

### 3.3.1 Sensitivity to shock volatility

The magnitude of the gap is governed by the variance of the shocks. Figure 3(b) shows that the total welfare difference rises steeply with volatility. Moving from $\sigma_Z = 0.2$ to the baseline 0.74 raises the shock variance by a factor of about 17 and the welfare difference by a factor of 16–20. At $\sigma_Z = 1$, the difference is 36–62 times its value at $\sigma_Z = 0.2$. Panel (a) reports the gap relative to the tax basket's welfare. In a two-species basket, it ranges from 0.1% at $\sigma_Z = 0.2$ to 4.7% at $\sigma_Z = 1$; at $N = 50$, it ranges from 0.004% to 0.14%. For every volatility, the relative gap declines in proportion to $1/N$, because the total difference stays flat while welfare grows with the number of species. The small gaps at low volatility therefore understate the quota's cost at empirically relevant volatility.

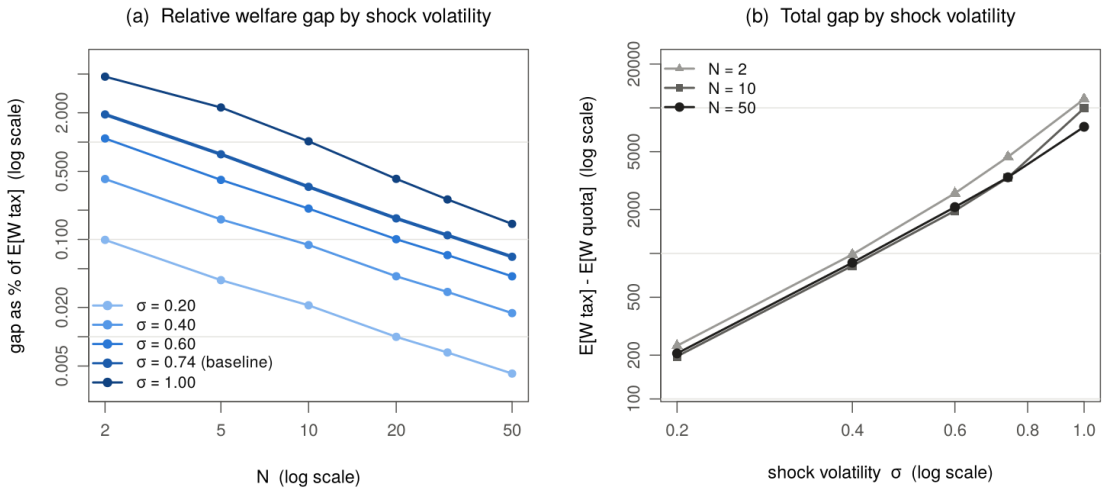



Figure 3: Sensitivity to shock volatility (balanced basket). Panel (a): welfare difference as a percentage of $\mathbb{E}[\mathcal{W}^\tau]$, log–log axes. The baseline $\sigma_Z = 0.74$ is the thick line. Panel (b): total welfare difference against $\sigma_Z$ for $N = 2, 10, 50$.

### 3.3.2 Decomposing the regulatory loss

Figure 4 shows that the two components of (5) move in opposite directions in $N$. The uncertainty loss is diversified away per species but not in total. It stays between 3,460 and 3,760 monetary units for $N \geq 5$ (4,428 at $N = 2$), and its per-species value falls from 2,214 at $N = 2$ to 70 at $N = 50$. The mismatch loss goes the opposite way. It grows roughly linearly, from 1,342 at $N = 2$ to 101,412 at $N = 50$, or 1.9% of first-best welfare.

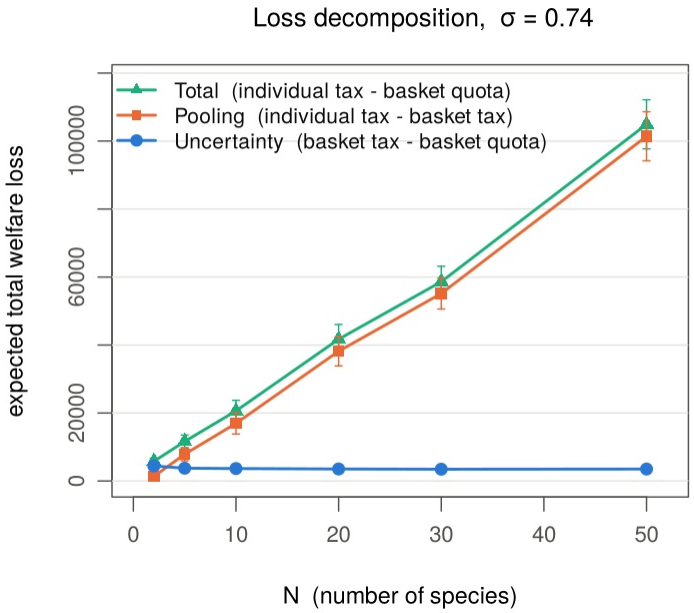



Figure 4: Decomposition of the total regulatory loss at $\sigma_Z = 0.74$, averaged over 25 balanced baskets per $N$ (300 shock paths each). The bars show $\pm 1.96$ standard errors across baskets.

The ranking of the two losses depends on basket size. In a two-species basket, the uncertainty loss is 3.3 times the mismatch loss (1.95% versus 0.59% of first-best welfare), so the choice between a tax and a quota is the first-order design decision. By $N = 5$, the mismatch loss is already twice the uncertainty loss, and by $N = 50$, it is 29 times larger. Using basket instruments therefore creates a trade-off between mismatch and uncertainty losses: the price–quantity choice matters most in small baskets, and the grouping decision dominates in large ones. This result extends previous research using groups of 2 (Clark, 1990), 5 (Collado et al., 2021), or 6 (Collado et al., 2024) species.

### 3.3.3 Robustness

With the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, Remark 1.2 expressed the per-species weights $\rho_t^i$ as a function of the escapements and cost parameters. Under a tax basket, each species reaches a constant escapement, and its weight also becomes constant. Under a quota basket, the weights change every period with $\lambda_t^F$ and stabilize only as $N \to \infty$. In both cases, the weights shrink as more species enter the basket. I define a basket as balanced when the largest weight also shrinks, $\max_i \rho^i \to 0$ as $N \to \infty$. What happens if one species dominates because it is highly valuable and abundant? Its weight then does not vanish, $\rho^i \to \bar{\rho} > 0$,

and the variance of its escapement,[5] $\mathrm{Var}(y_t^i) \approx (\rho^i)^2 \mathrm{Var}(S_t) = \bar{\rho}^2\,\Theta(N)$, grows with $N$. In other words, a weight that does not shrink prevents the basket from spreading shocks across species.

To test this, I hold one species' share of aggregate biomass fixed at 60% as $N$ grows, so its shock never diversifies away, and repeat the convergence exercise (one basket per $N$) with this dominant species. The per-species difference no longer vanishes. It stays between about 2,600 and 3,800 for $N \geq 5$, and the total difference grows from 3,165 at $N = 2$ to 147,014 at $N = 50$.

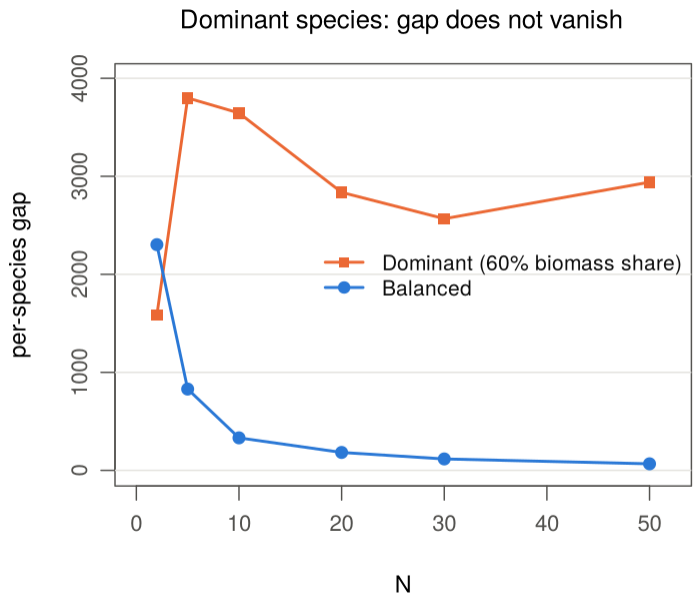



Figure 5: Robustness with one persistently dominant species. The per-species welfare difference does not vanish as basket size increases, unlike in a balanced basket.

This case shows that bundling does not work for every species composition: a successful basket is a balanced one, because balance guarantees diversification. I conjecture that relaxing constant catchability would blur the tax basket's advantage over the quota but leave the mismatch loss intact. Appendix B.3 shows only that the tax condition then becomes a lower bound.

A second check has each species draw its own volatility, $\sigma_Z^i \sim \mathcal{U}[0.40, 1.08]$. This range centers on the baseline 0.74 and spans about one predictive standard deviation (0.35) on each side in Thorson et al.'s stock-level estimates (2014). For each $N$, I keep the main exercise basket balanced and average over 10 draws of the volatility vector, with 400

[5] Under the quota, a species absorbs the fraction $\rho_t^i$ of the aggregate surprise, $y_t^i - \bar{y}^i \approx \rho_t^i S_t$, by (55).

shock paths per draw. The total welfare difference is 12–41% larger than with a common $\sigma_Z = 0.74$. Because the shock variance is convex in $\sigma$ (for the untruncated lognormal, $\mathrm{Var}(Z) = e^{\sigma^2} - 1$), spreading volatilities around 0.74 raises the average variance. The per-species difference still declines at rate $1/N$, from 2,679 at $N = 2$ to 75 at $N = 50$ (panel (b) of Figure 6). Diversification therefore does not require species to share the same volatility.

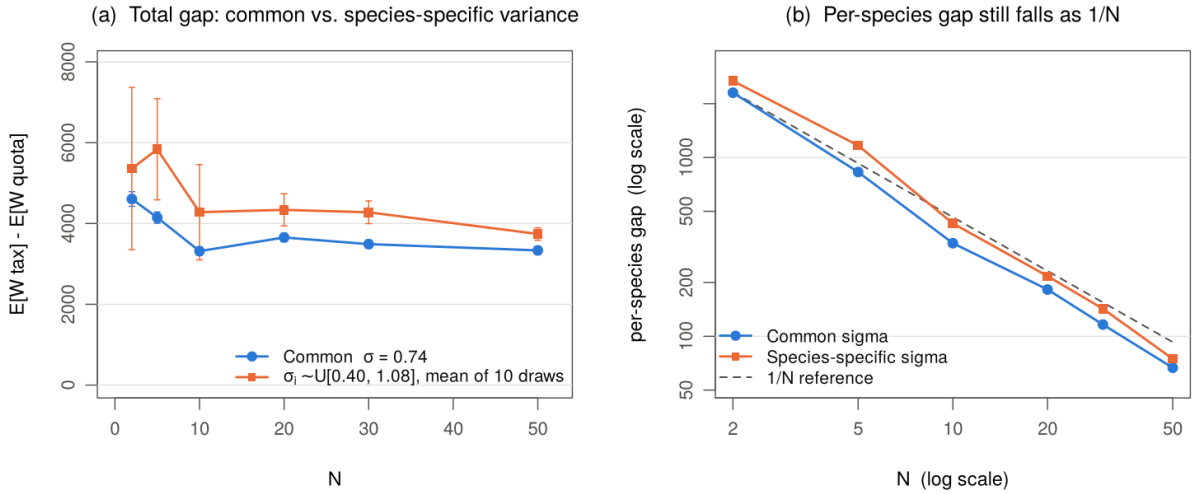



Figure 6: Robustness to species-specific volatilities, $\sigma_Z^i \sim \mathcal{U}[0.40, 1.08]$ (mean of 10 draws; the bars are $\pm 1.96$ standard errors across draws), against a common $\sigma_Z = 0.74$ on the same basket. Panel (a): total welfare difference. Panel (b): per-species difference on log–log axes with a $1/N$ reference.

# 4 Discussion and conclusion

This is the first paper to study the incentives and trade-offs of basket instruments for heterogeneous stocks under ecological uncertainty. Building on and extending the seminal work of Reed (1979) and Weitzman (2002), I find that tax baskets outperform quota baskets in welfare. However, the welfare gap between instruments narrows to a constant as more species are included in the bundle. As the basket grows, a trade-off emerges between diversifying the uncertainty loss and increasing the mismatch loss relative to a first-best policy. Whether the policymaker uses a tax basket or a quota basket, fishermen keep fishing each period until marginal profits are equalized across all fishable species. This margin integrates biological productivity, harvesting costs, and market prices. These findings help unify a literature that proposes grouping species based on similar ecological (Costello and Szuwalski, 2018), economic (Collado et al., 2021), or vulnerability (Collado et al., 2024) parameters.

In general, both policies are second-best instruments, and tax baskets outperform quota baskets in welfare. Both target a common margin equal to a weighted average of the species' next-period marginal profits, analogous to the optimal uniform tax across multiple pollutants (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). Because the problem is dynamic, the weights can change every period. In this context, a tax basket is particularly helpful because it keeps the remaining biomass (and the weights) constant over time and is immune to ecological shocks. Nevertheless, the tax basket falls further behind species-specific instruments as $N$ increases, because a single tax cannot match many different species-specific optimal taxes. A tax basket achieves the first-best only under a specific condition: each species' marginal profit at its own Reed escapement must be the same (equivalently, all individual optimal taxes must coincide). Identical species, or species with very similar economic and biological parameters as Kritzer et al. (2023) recommend, are a special case. By duality, the same condition also makes quota baskets first-best in a deterministic setting.

Quota baskets, by contrast, carry an additional distortion under uncertainty. The timing asymmetry between manager and fisherman does not affect the tax basket, because the tax induces the fisherman to stop harvesting when the stock reaches its target escapement, whatever the shock. When the quota basket includes many species and ecological shocks are independent, however, shocks partly average out, reducing the per-species loss relative to the tax basket. In this case, quota baskets approximate the second-best tax on a per-species basis, and the total welfare gap converges to a positive constant as $N$ increases. Under the model's assumptions, prices still dominate quantities, but diversification narrows the per-species gap. Finally, as basket size increases, the quota basket also falls further behind species-specific instruments.

The trade-off between uncertainty diversification and policy mismatch makes the choice of $N$ a central design decision. In the simulations, at empirically calibrated recruitment volatility, the tax outperforms the quota by about 2% of the tax basket's welfare in a

two-species basket but by less than 0.1% at $N = 50$, while the mismatch loss grows to about 1.9% of first-best welfare at $N = 50$. While this paper does not determine the optimal basket size, it shows that composition matters. The planner must decide whether the administrative savings justify the mismatch loss. If monitoring capabilities are limited, a bundle instrument is recommended as long as the total welfare loss does not exceed the regulatory cost savings. A larger basket is not always better.

On the other hand, if institutional and scientific capabilities are well developed, a large basket may not be a priority, because monitoring one additional species can be relatively cheap when the number of monitored species is already high. This may explain the stark differences in multispecies management (California, China, Alaska, Chile, and Thailand). When datasets are hard to build, managers struggle to distinguish species, or fishing gear is not selective enough, a basket instrument can be attractive. Basket instruments can complement the super-species models (pooling all species into a single stock) used in Southeast Asia (Kulanujaree et al., 2020). However, it does not address system-wide shifts such as climate change (Fulton et al., 2022) because shocks may not cancel out.

These findings are particularly relevant for fisheries characterized by low selectivity, limited data, and high landing aggregation. For example, China harvests many species jointly, and regulators have limited ability to identify or regulate individual stocks (Szuwalski et al., 2017). China's summer fishing moratorium, a seasonal closure applied to essentially all vessels and gear across the Bohai, Yellow, East, and South China Seas (Szuwalski et al., 2017), functions as a whole-fishery effort control, the most aggregated basket possible. Although the model studies landing fees and quotas rather than effort controls, its logic suggests that such an instrument smooths ecological variability across stocks while incurring a mismatch loss that rises with the number of species swept into it. This reasoning aligns with broader evidence that diversification across multiple stocks can reduce income volatility and targeting risk for fishers (Cline et al., 2017; Robinson et al., 2020). In contrast, aggregate production models have proven tractable in fisheries such as Thailand's, but doubts remain about their resilience under ecological uncertainty (Fulton et al., 2022). Conceptually, the result parallels findings in environmental economics showing that uniform instruments can perform well when uncertainty is diversified across regulated sources (Fowlie and Muller, 2019).

Few countries employ landing fees, and where they exist, they typically complement rather than replace quota systems (Ben-Hasan et al., 2021), despite Weitzman's (2002) case for prices over quantities. This paper does not overturn that ranking: under ecological uncertainty, the tax basket remains the welfare-preferred instrument. Yet if tax baskets are politically infeasible, this paper provides conditions under which a quota basket can narrow the welfare gap.

For the policymaker, basket instruments are most attractive in data-limited settings where species-specific regulation is infeasible. Tax baskets do not require real-time stock information at the start of period $t$ or during it. While they reduce monitoring and species-identification costs, harvest-cost parameters are not readily available for all fisheries and

may be difficult for authorities to obtain. By duality, quota baskets benefit from aggregation but require periodic updates and reliable aggregate biomass information. This weakens the presumption that basket regulation reduces monitoring costs. Indicator species may partially alleviate this burden, but they introduce additional measurement error that compounds ecological uncertainty and further degrades quota performance.[6] These distinctions suggest that the choice between instruments should depend not only on ecological uncertainty but also on the regulator's institutional and budgetary capacity.

Several limitations point to directions for future research. First, the analysis abstracts from fishing technology choice and gear selectivity. Incorporating endogenous targeting decisions could clarify when basket instruments induce efficiency losses or gains through technological adjustment. Second, the framework focuses on a single basket; extending it to multiple, overlapping baskets raises questions about species mobility across regulatory groups, enforcement, and permit trade and pricing. Third, the model does not represent species interactions, and preliminary derivations suggest that adding them makes the ranking of instruments ambiguous. Fourth, linking the marginal-profit condition to trait-based or size-spectrum ecosystem models (Andersen et al., 2016) could help translate the theoretical criteria into empirically observable metrics, informing practical basket design that balances efficiency, biodiversity, and risk. Fifth, some results rest on the form of the cost function, $C^i(s^i) = \frac{\theta^i}{s^i}$, which makes the tax basket stock-independent. Relaxing it to non-constant catchability would likely narrow the tax's advantage.

Finally, this work does not consider bycatch of low-value or overfished species (e.g., canary and yelloweye rockfish on the U.S. West Coast). A bundle instrument could delay the recovery of vulnerable species, but it would not heavily restrict the extraction of healthier stocks. However, an unhealthy stock is problematic under the model's differentiated costs and fishable-set assumption. With differentiated costs, the fisherman decides what to catch, and the regularity condition fails if a stock becomes scarce after a shock. If species enter and exit the fishable set, the outcome depends on the instrument. The simulations show that a species' harvest drops to zero when it is no longer profitable after a negative shock; in that case, the fisherman can compensate by catching other species. In a tax basket, a tax set to include a non-fishable species distorts the margins of the other species. Ideally, the tax would be less stringent and target only the participating species. Thus, the manager must invest in information on participating species, because ecological shocks may change species participation after the policy is set. This creates a trade-off between conservation (designing the policy to include all species) and profit maximization (designing the policy to include only potentially fishable species). In short, bundle instruments cannot fully resolve the sustainability–profitability trade-off they were created to address.

In summary, tax baskets are generally more robust under uncertainty, and both instruments can perform well in highly aggregated systems or under restrictive conditions.

[6] Sethi et al. (2005) demonstrated that measurement uncertainty has the greatest impact on fishery policy, profits, and fishery collapse under single-species management.

However, managing larger baskets involves a trade-off among diversifying uncertainty loss, increasing mismatch loss, and reducing administrative costs. Bundling does not replace protection for vulnerable, non-target stocks. However, with careful attention to information requirements and ecological risk, basket regulation offers a pragmatic complement to species-specific management. Finally, implementing a basket instrument can be a stepping stone toward more stringent regulation.

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# Appendix A: Derivation of the optimal tax basket (1 species)

## A.1: Fisherman's problem

The myopic fisherman maximizes profits by harvesting fish in period $t$:

$$\pi_t(x) = \max_{h_t \geq 0} \left\{ \int_{x_t - h_t}^{x_t} (p - \tau_t - C(s))\, ds \right\} \tag{27}$$

where $p$ is the unit price of fish, $C(s)$ is the unit cost of harvesting at fish population $s$, and $\tau_t$ is the landing fee per unit of fish. Taking the first-order condition (FOC) with respect to harvest, I obtain the optimality condition for the current period:

$$p - C\left(x_t - h_t\right) = \tau_t$$

Using the definition of escapement $y_t = x_t - h_t$, I can re-express this condition as:

$$p - C\left(y_t\right) = \tau_t \tag{28}$$

The fisherman therefore makes an escapement decision:

$$\begin{aligned} p - \tau_t - C(y_t) &= 0 \\ C(y_t) &= p - \tau_t \\ y_t &= C^{-1}(p - \tau_t) \end{aligned} \tag{29}$$

## A.2: Social planner's problem

The social planner picks a landing fee to induce harvest levels that maximize expected present-discounted profits:

$$\mathbb{E}\Big[\sum_{t=0}^{\infty} \beta^t \int_{X_t - h_t}^{X_t} \pi(s)\, ds\Big] \tag{30}$$

where $\beta \equiv \frac{1}{1+\delta} \in (0,1)$, $\delta$ is the discount rate, and the stock at the start of period $t$ is $X_t = Z_t G(y_{t-1})$.

## A.3: Landing fee

The manager sets $\tau_t$ taking into account the fisherman's decision function, with the previous end-of-period escapement $C^{-1}(p - \tau_{t-1})$ as the relevant state variable:

$$V(Y) = \max_{\tau \geq 0} \mathbb{E}\left[ \int_{C^{-1}(p-\tau_t)}^{Z_t G(C^{-1}(p-\tau_{t-1}))} \pi(s)\, ds + \beta V(C^{-1}(p - \tau_t)) \right] \tag{31}$$

The profit reduction from the tax and the tax revenue cancel each other. With this, I take the first-order condition with respect to $\tau_t$:

$$\frac{\partial V(.)}{\partial \tau_t} = \mathbb{E}\left[-\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] + \beta\mathbb{E}\left[\frac{d}{d\tau_t}[V_{t+1}(.)]\right] = 0 \tag{32}$$

Then, differentiating with respect to $\tau_{t-1}$ and using the Leibniz integral rule:

$$\frac{\partial V_t(.)}{\partial \tau_{t-1}} = \mathbb{E}\left[\left(p - C(Z_t G(y_{t-1}(\tau_{t-1})))\right)\frac{d}{d\tau_{t-1}}\left(Z_t G(y_{t-1}(\tau_{t-1}))\right)\right] \tag{33}$$

By the Benveniste–Scheinkman theorem, I move expression (33) forward by one period:

$$\frac{\partial V_{t+1}(.)}{\partial \tau_t} = \mathbb{E}\left[\left(p - C(Z_{t+1} G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1} G(y_t(\tau_t))\right)\right]$$

Putting everything together:

$$\begin{aligned}
\mathbb{E}\left[\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\mathbb{E}\left[\left(p - C(C^{-1}(p-\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \text{ Replace } y_t(\tau_t) \\
\mathbb{E}\left[\left(p - (p-\tau_t)\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\mathbb{E}\left[\tau_t\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[\frac{dZ_{t+1}}{d\tau_t}G(y_t(\tau_t)) + Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\right]\right]\frac{dy_t}{d\tau_t} \\
\tau_t &= \beta\mathbb{E}\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t} - C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] \\
\tau_t &= \beta\mathbb{E}\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] - \beta\mathbb{E}\left[C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right] \\
\tau_t &= \beta p\frac{dG(y_t)}{dy_t} - \beta\mathbb{E}\left[Z_{t+1}C(Z_{t+1}G(y_t))\right]\frac{dG(y_t)}{dy_t}
\end{aligned}$$

In the derivation above, the expression is conditioned on period-$t$ information, and taxes do not influence shocks.[7]

[7] A more stringent assumption is independence between $Z$, $G$, and $C$.

To make the result comparable with Reed (1979), I employ a cost function of the family $C(s) = \frac{\theta}{s^\alpha}$. This function is homogeneous: $C(ms) = \frac{\theta}{(ms)^\alpha} = m^{-\alpha}\frac{\theta}{s^\alpha}$. The expression above can then be written in two ways, depending on the catchability assumption:

$$\begin{aligned}
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta \mathbb{E}\left[Z_{t+1} C(Z_{t+1} G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta \mathbb{E}\left[Z_{t+1} Z_{t+1}^{-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta \mathbb{E}\left[Z_{t+1}^{1-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta \left[p - \mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t}
\end{aligned}$$

For the term $\mathbb{E}\left[Z_{t+1}^{1-\alpha}\right]$, there are two cases:

- With constant catchability, $\alpha = 1$, $\mathbb{E}\left[Z_{t+1}^{1-1}\right] = 1$, and the tax does not depend on the current stocks: $\tau_t = \beta \left[p - (C \circ G)(y_t)\right] \frac{dG(y_t)}{dy_t}$.
- With decreasing catchability, $0 < \alpha < 1$, Jensen's inequality gives $\mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] \leq \mathbb{E}\left[Z_{t+1}\right]^{1-\alpha} = 1$. Hence, $\tau_t \geq \beta \left[p - (C \circ G)(y_t)\right] \frac{dG(y_t)}{dy_t}$: the deterministic expression is a lower bound (a minimum tax).

The right-hand side is part of the Euler equation for constant escapement proposed by Clark (1976). It is then possible to propose a constant tax based on the constant optimal escapement $y^*$ that satisfies the Euler equation:

$$\begin{aligned}
\tau_t &= \beta \left[p - \mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t} = (p - C(y_t)) \\
\tau^* &= \beta \left[p - \mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] C(G(y^*))\right] \frac{dG(y^*)}{dy^*} = (p - C(y^*)) \\
\tau^* &= \pi(y^*)
\end{aligned}$$

The optimal tax depends only on the optimal escapement, as Weitzman (2002) shows.

## Appendix B: Derivation of the optimal marginal profit $m\pi$

### B.1: Fisherman's problem

In each period $t$, a myopic fisherman maximizes profits by harvesting $N$ stocks $i = 1, 2, \ldots, N$ with independent life cycles, given that the planner uses an instrument that sets a target $m\pi_t$:

$$\begin{aligned} \pi_t = \max_{\{h_t^i\}_{i=1}^N} & \left\{ \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} \left(p^i - m\pi_t - C^i(s^i)\right) ds^i \right\} \\ \text{s.t. } & h_t^i \geq 0 \ \forall i \end{aligned} \tag{34}$$

where $m\pi_t$ is the target marginal profit, which can be implemented through a landing fee per unit of fish or a shared quota applied to all species. From the FOC with respect to each harvest, I obtain the optimality condition for the current period:

$$p^1 - C^1\left(X_t^1 - h_t^1\right) = p^2 - C^2\left(X_t^2 - h_t^2\right) = \cdots = p^N - C^N\left(X_t^N - h_t^N\right) = m\pi_t$$

Using the definition of escapement $y_t^i = X_t^i - h_t^i$, I can re-express this condition as:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = m\pi_t \tag{35}$$

Equation (35) is the equimarginal principle. As in the previous section, the fisherman's escapement decision is:

$$\begin{aligned} p^i - m\pi_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - m\pi_t \\ y_t^i &= (C^i)^{-1}(p^i - m\pi_t) \end{aligned} \tag{36}$$

For a given margin $m\pi_t$, the fisherman's escapement decision is independent of stock size and, therefore, unaffected by the ecological shocks.

### B.2: Social planner's problem

The social planner incorporates the fisherman's escapement decision into her dynamic optimization problem:

$$V(\vec{y}_{t-1}) = \max_{m\pi_t \geq 0} \ \mathbb{E}\left\{ \sum_{i=1}^{N} \int_{Y^i(m\pi_t)}^{Z_t^i G^i(y_{t-1}^i)} \left(p^i - C^i(s^i)\right) ds^i + \beta V(\vec{y}_t) \right\} \tag{37}$$

The target marginal profit depends on the available information. If the manager does not know the current stock status, I assume she operates based on each $y_{t-1}^i$. Although the

planner cannot observe current ecological shocks, I include them in the dynamic problem to evaluate their impact on the tax basket. I take the FOC with respect to $m\pi_t$ (following the same procedure as in Appendix A) and obtain the following Euler equation:

$$\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y_t^i))\, y_{m\pi_t}^i\Big] = \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big]$$

Replacing the fisherman's condition $y_t^i = (C^i)^{-1}(p^i - m\pi_t)$:

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y_t^i))\, y_{m\pi_t}^i\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i((C^i)^{-1}(p^i - m\pi_t))\big)\, y_{m\pi_t}^i\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - (p^i - m\pi_t)\big)\, y_{m\pi_t}^i\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big] \\
\sum_{i=1}^{N} m\pi_t\, y_{m\pi_t}^i &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big] \\
m\pi_t \sum_{i=1}^{N} y_{m\pi_t}^i &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big] \\
m\pi_t^+ &= \beta\frac{1}{\sum_{j=1}^{N} y_{m\pi_t}^j}\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)\, y_{m\pi_t}^i\Big] \\
m\pi_t^+ &= \sum_{i=1}^{N}\mathbb{E}\big[\pi_{t+1}^i\big]\frac{y_{m\pi_t}^i}{\sum_{j=1}^{N} y_{m\pi_t}^j}
\end{aligned} \tag{38}$$

If the cost function is $C^i(s^i) = \frac{\theta^i}{s^i}$, then $y_{m\pi_t}^i = \frac{\theta^i}{(p^i - m\pi_t)^2}$:

$$\begin{aligned}
m\pi_t^+ &= \sum_{i=1}^{N}\mathbb{E}\big[\pi_{t+1}^i\big]\frac{y_{m\pi_t}^i}{\sum_{j=1}^{N} y_{m\pi_t}^j} \\
&= \sum_{i=1}^{N}\mathbb{E}\big[\pi_{t+1}^i\big]\frac{\frac{\theta^i}{(p^i - m\pi_t)^2}}{\sum_{j=1}^{N}\frac{\theta^j}{(p^j - m\pi_t)^2}}
\end{aligned} \tag{39}$$

The fisherman's decision is $p^i - m\pi_t = C^i(y_t^i)$, with $C^i(y_t^i) = \frac{\theta^i}{y_t^i}$:

$$
\begin{aligned}
m\pi_t^+ &= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \frac{\frac{\theta^i}{(p^i - m\pi_t)^2}}{\sum_{j=1}^{N} \frac{\theta^j}{(p^j - m\pi_t)^2}} \\
&= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \frac{\frac{\theta^i}{C^i(y_t^i)^2}}{\sum_{j=1}^{N} \frac{\theta^j}{C^j(y_t^j)^2}} \\
&= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \frac{\frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}}{\sum_{j=1}^{N} \frac{\left(Y^j(m\pi^+)\right)^2}{\theta^j}} \\
&= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \rho_t^i
\end{aligned}
\tag{40}
$$

The weight of each species depends on its escapement (remaining biomass) and its cost parameter.

### B.3: Tax basket equivalence

Replacing $m\pi_t = \tau_t$, it is straightforward to obtain the same expression as in Appendix B.2:

$$
\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p^i - \mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right] C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}
$$

For the term $\mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right]$, there are two cases:

- With constant catchability, $\alpha = 1$, $\mathbb{E}\left[(Z_{t+1}^i)^0\right] = 1$, so the tax does not depend on the current stocks.
- With decreasing catchability, $0 < \alpha < 1$, Jensen's inequality gives $\mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right] \leq \mathbb{E}\left[Z_{t+1}^i\right]^{1-\alpha} = 1$. Hence, the deterministic expression for $\tau_t$ is a lower bound (a minimum tax).

The tax basket is the weighted average of the species' marginal profits in the next period. Assuming $\alpha = 1$ and the cost function $C^i = \frac{\theta^i}{s^i}$ gives:

- $y_t^i = \frac{\theta^i}{p^i - \tau_t}$
- $\frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$

From the fisherman's optimization, $p^i - \tau_t = C^i(y_t^i)$ and $C^i(y_t^i) = \frac{\theta^i}{y_t^i}$. With this information, I can rearrange the tax basket:

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p^i - \mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right] C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t} \qquad \text{Replace } \alpha = 1$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{(p^i-\tau_t)^2}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{\theta^i}{(p^i-\tau_t)^2} \qquad \text{Replace } \frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i-\tau_t)^2}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{C^i(y_t^i)^2}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{\theta^i}{C^i(y_t^i)^2} \qquad \text{Replace } p^i - \tau_t = C^i(y_t^i)$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i)^2}{\theta^i}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{(y_t^i)^2}{\theta^i} \qquad \text{Replace } C^i(y_t^i) = \frac{\theta^i}{y_t^i}$$

$$\tau_t = \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i(\tau_t))^2}{\theta^i}} \sum_{i=1}^{N} \pi_{t+1}^i(y_t^i(\tau_t)) \frac{(y_t^i(\tau_t))^2}{\theta^i} \qquad \text{Rename next period's marginal profit}$$

$$\tau_t = \tau_t(\vec{y}_t)$$

As with $m\pi_t$, the tax basket is a function of each species' escapement. If the species are identical, the tax basket equals the landing fee for an individual species, as shown in Appendix A. Moreover, the social planner does not need to know the ecological shocks in the current period.

# Appendix C: Auxiliary proofs

## Appendix C.1: Welfare rearrangement

By Lemma 1, the fisherman equates $p^i - C^i(y_t^i) = m\pi_t$, with $m\pi_t = \tau_t$ under the tax and $m\pi_t = \lambda_t$ under the quota, so $y_t^i = Y^i(m\pi_t)$. If $X_t^i > y_t^i$ for all $i$, the lemma holds every period. Therefore, escapement depends on the ecological shocks only through the common margin. With $C^i(s) = \theta^i/s$, the fisherman's myopic payoff is

$$\int_{y_t^i}^{X_t^i} \Big(p^i - \frac{\theta^i}{s}\Big) ds = A^i(X_t^i) + \phi^i(y_t^i), \quad \text{where } A^i(X) := p^i X - \theta^i \ln X, \quad \phi^i(y) := -p^i y + \theta^i \ln y,$$

which is (15). The current stock and the current escapement enter additively. Substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, and using $\mathbb{E}[Z_t^i] = 1$, $\mathbb{E}|\ln Z_t^i| < \infty$, and $\ln(Z_t^i G^i) = \ln Z_t^i + \ln G^i$,

$$\begin{aligned}
\mathbb{E}\big[A^i\big(Z_t^i G^i(y_{t-1}^i)\big)\big] &= \mathbb{E}\big[p^i X - \theta^i \ln X\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln(Z_t^i G^i)\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln G^i - \theta^i \ln Z_t^i\big] \\
&= p^i G^i(y_{t-1}^i) - \theta^i \ln G^i(y_{t-1}^i) - \theta^i \,\mathbb{E}[\ln Z_t^i] \\
&= a^i(y_{t-1}^i) - \kappa^i
\end{aligned}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$, $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$, and $|\mathbb{E}[\ln Z_t^i]| < \infty$ by assumption. Then, taking the initial stock $X_0^i$ (and $y_0^i$) as given and summing across species,

$$\mathcal{W} = \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] \Big\}$$

Shifting the index in $\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i) = \beta \sum_{t\geq 0} \beta^t a^i(y_t^i)$ and reallocating $\beta^t$,

$$\begin{aligned}
\sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] &= \underbrace{\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i)}_{\text{(I): stock block}} - \underbrace{\sum_{t\geq 1} \beta^t \kappa^i}_{\text{(II): shock block}} + \underbrace{\sum_{t\geq 1} \beta^t \phi^i(y_t^i)}_{\text{(III): escapement block}} \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \sum_{t\geq 1} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \beta \sum_{t\geq 0} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i)
\end{aligned}$$

Putting things together:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \Big\} \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_i \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_i \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \sum_i \phi^i(y_t^i) \\
&= \sum_i \Big[ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} \Big] + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) + \sum_{t\geq 0} \beta^t \sum_i \phi^i(y_t^i) \\
&= \underbrace{\sum_i \Big[ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} \Big]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \big[ \underbrace{\phi^i(y_t^i) + \beta a^i(y_t^i)}_{=:w^i(y_t^i)} \big] \\
&= \Xi_0 + \sum_{t\geq 0} \beta^t \sum_i w^i(y_t^i)
\end{aligned}
\tag{41}
$$

This result applies to both the tax and the quota basket.

### Appendix C.2: Recovering the dynamic solution through the static problem

Returning to equation (41) and assuming that each $y_t^i = y^i$ is constant over time:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[ \phi^i(y^i) + \beta a^i(y^i) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[ \beta \big( p^i G^i(y^i) - \theta^i \ln G^i(y^i) \big) - \big( p^i y^i - \theta^i \ln y^i \big) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \underbrace{\frac{1}{1-\beta} \Big[ \beta p^i G^i(y^i) - \beta \theta^i \ln G^i(y^i) - p^i y^i + \theta^i \ln y^i \Big]}_{\mathcal{V}^i} \Big\}
\end{aligned}
\tag{42}
$$

I take the derivative with respect to each $y^i$:

$$
\begin{aligned}
\frac{\partial \mathcal{W}}{\partial y^i} &= \frac{1}{1-\beta}\Big[\beta p^i G^{i\prime}(y^i) - \beta\theta^i \frac{1}{G^i(y^i)} G^{i\prime}(y^i) - p^i + \theta^i \frac{1}{y^i}\Big] \\
&= \frac{1}{1-\beta}\Big[\beta\Big(p^i - \theta^i \frac{1}{G^i(y^i)}\Big) G^{i\prime}(y^i) - \Big(p^i - \theta^i \frac{1}{y^i}\Big)\Big] \\
&= \frac{1}{1-\beta}\Big[\beta\big(p^i - C^i(G^i(y^i))\big) G^{i\prime}(y^i) - \big(p^i - C^i(y^i)\big)\Big] \\
&= \frac{1}{1-\beta}\Big[\pi^i_{t+1}(y^i) - \big(p^i - C^i(y^i)\big)\Big]
\end{aligned}
\tag{43}
$$

Setting (43) to zero recovers Reed's condition for species $i$; imposing a common margin $y^i = Y^i(\tau)$ and weighting by $dY^i/d\tau$ recovers the Euler equation of Proposition 2. For species-specific management, the discount factor multiplies the growth block in (42), which is what makes (43) vanish exactly at Reed's $y^{i*}$. Because $\pi^{i\prime}_{t+1} \leq 0$ and $C^{i\prime} < 0$, $\mathcal{V}^{i\prime\prime} < 0$: $\mathcal{V}^i$ is strictly concave with unique maximizer $y^{i*}$. Hence,

$$
\pi^i_{t+1}(y^i) > p^i - C^i(y^i) \;\text{ for } y^i < y^{i*}, \qquad \pi^i_{t+1}(y^i) < p^i - C^i(y^i) \;\text{ for } y^i > y^{i*} \tag{44}
$$

This sign rule is key to proving uniqueness.

## Appendix D: Existence, uniqueness and interiority of $\tau^+$

Given Proposition 2 and Appendix C.2, the regularity condition $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$, and the cost function $C^i = \theta^i/s^i$, I can rewrite the tax optimality condition as a residual function:

$$g(\tau) := \tau - R(\tau), \qquad \text{where } R(\tau) := \frac{\sum_i \pi^i(Y^i(\tau))\,(Y^i(\tau))^2/\theta^i}{\sum_i (Y^i(\tau))^2/\theta^i}.$$

Since $\mathcal{W}'(\tau) = -\big(\sum_i Y^{i\prime}(\tau)\big)g(\tau)$, $g$ and $\mathcal{W}'$ have the same roots and opposite signs.

### Appendix D.1: Endpoint signs

From Appendix C.2, the tax solution commits species $i$ to a constant escapement $y$ forever. Without loss of generality, order species by their individual optimal taxes, $\tau^{1*} \le \cdots \le \tau^{N*}$, and assume not all $\tau^{i*}$ are equal. Then:

1. $Y^i(\cdot)$ is strictly increasing, so $\tau < \tau^{i*} \iff Y^i(\tau) < y^{i*}$ (a lower tax leaves less stock in the ocean).
2. At $\tau = \tau^{1*}$, species 1 contributes exactly $\pi^1_{t+1}(Y^1(\tau^{1*})) = \tau^{1*}$ (shown in Appendix A).
3. Each $j \ge 2$ has $\tau^{1*} < \tau^{j*}$, hence $Y^j(\tau^{1*}) < y^{j*}$ (overharvest), and (44) with $p^j - C^j(Y^j(\tau^{1*})) = \tau^{1*}$ gives $\pi^j_{t+1}(Y^j(\tau^{1*})) > \tau^{1*}$.
4. A weighted average with strictly positive weights (guaranteed by the fishable range) of terms that are all $\ge \tau^{1*}$, at least one strictly, exceeds $\tau^{1*}$. Therefore $R(\tau^{1*}) > \tau^{1*}$ and $g(\tau^{1*}) < 0$.
5. At the opposite endpoint, $\tau = \tau^{N*}$, each $j < N$ symmetrically has $Y^j(\tau^{N*}) > y^{j*}$ (underharvest), so $\pi^j_{t+1}(Y^j(\tau^{N*})) < \tau^{N*}$, giving $R(\tau^{N*}) < \tau^{N*}$ and $g(\tau^{N*}) > 0$.

### Appendix D.2: Existence

$Y^i$, $\pi^i$, and the denominator $\sum_i \theta^i/(p^i - \tau)^2$ are continuous on $[\tau^{1*}, \tau^{N*}]$. The interior regularity condition guarantees $p^i - \tau \ge C^i(K^i) > 0$; the denominator is then bounded and bounded away from zero, and $g$ is continuous. By the Intermediate Value Theorem, there exists $\tau^+ \in (\tau^{1*}, \tau^{N*})$ with $g(\tau^+) = 0$.

### Appendix D.3: Uniqueness

A sufficient condition is that $\mathcal{W}$ be strictly concave in $\tau$ on $[\tau^{1*}, \tau^{N*}]$. Concavity of each $w^i$ in its own escapement (Appendix C.2) is not enough, because $\tau$ enters $w^i$ through $Y^i$,

and $Y^{i\prime}$ sets the weights. Without loss of generality, I focus on $W(\tau) = \sum_i w^i(Y^i(\tau))$. Differentiating $W$ twice:

$$W''(\tau) = \sum_i \Big[\underbrace{w^{i\prime\prime}(y^i)\left(Y^{i\prime}(\tau)\right)^2}_{\leq 0 \text{ if } w^i \text{ concave}} + \underbrace{w^{i\prime}(y^i)\,Y^{i\prime\prime}(\tau)}_{\text{indefinite}}\Big], \qquad y^i = Y^i(\tau) \tag{45}$$

The $y^i$ differ across species even though $\tau$ is shared, and the sum contains no cross terms ($W$ is separable). Per-species concavity is therefore sufficient for global concavity. However, the term $w^{i\prime}(y^i)\,Y^{i\prime\prime}(\tau)$ has an ambiguous sign: $Y^{i\prime\prime}(\tau) = 2\theta^i/(p^i-\tau)^3 > 0$, while $w^{i\prime}(y^i) = \pi^i_{t+1}(y^i) - \tau$ is positive below $\tau^{i*}$ and negative above it by (44).

To handle this, I remove the nonlinearity of $Y^i$ through an affine reparametrization. From the fisherman's rule $p^i - \theta^i/y^i = \tau$,

$$p^i - \frac{\theta^i}{y^i} = \tau$$
$$z^i := \frac{1}{y^i} = \frac{p^i - \tau}{\theta^i}$$

is affine in $\tau$, so $z^{i\prime\prime}(\tau) = 0$ and the curvature term vanishes. I rewrite $W$:

$$W(\tau) = \sum_i w^i(y^i) = \sum_i w^i(1/z^i) = \sum_i \hat{w}^i(z^i), \quad \text{where } \hat{w}^i(z) := w^i(1/z).$$

Because each $z^i$ is affine in $\tau$ and $W$ is separable, $W''(\tau) = \sum_i \hat{w}^{i\prime\prime}(z^i)\,(z^{i\prime})^2$ with $(z^{i\prime})^2 > 0$. Hence $W$ is concave in $\tau$ if $\hat{w}^{i\prime\prime}(z^i) \leq 0$ for all $i$, and strictly concave if, in addition, the inequality is strict for at least one $i$. Using $\dfrac{dy}{dz} = \dfrac{d}{dz}z^{-1} = -\dfrac{1}{z^2}$ and $z = 1/y \Rightarrow \dfrac{1}{z^2} = y^2$,

$$\hat{w}^{i\prime}(z) = w^{i\prime}\left(\tfrac{1}{z}\right)\left(-\tfrac{1}{z^2}\right) = -\tfrac{1}{z^2}\,w^{i\prime}(y)$$

and

$$\hat{w}^{i\prime\prime}(z) = \tfrac{2}{z^3}w^{i\prime}\left(\tfrac{1}{z}\right) - \tfrac{1}{z^2}w^{i\prime\prime}\left(\tfrac{1}{z}\right)\left(-\tfrac{1}{z^2}\right) = \tfrac{2}{z^3}w^{i\prime}\left(\tfrac{1}{z}\right) + \tfrac{1}{z^4}w^{i\prime\prime}\left(\tfrac{1}{z}\right).$$

Replacing $z = 1/y$ (so $1/z^3 = y^3$, $1/z^4 = y^4$),

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= 2y^3 w^{i\prime}(y) + y^4 w^{i\prime\prime}(y)\\ &= y^3\left[\,y\,w^{i\prime\prime}(y) + 2w^{i\prime}(y)\,\right]\end{aligned}$$

From Appendix C.2, $w^{i\prime}(y) = \pi^i_{t+1}(y) - p^i + \theta^i/y$ and $w^{i\prime\prime}(y) = \pi^{i\prime}_{t+1}(y) - \theta^i/y^2$, so

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= y^3\left[\,y\,(\pi^{i\prime}_{t+1}(y) - \theta^i/y^2) + 2(\pi^i_{t+1}(y) - p^i + \theta^i/y)\right]\\ &= y^3\left[y\pi^{i\prime}_{t+1}(y) - \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i + 2\theta^i/y\right]\\ &= y^3\left[y\pi^{i\prime}_{t+1}(y) + \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i\right]\\ &= y^3\left[y\pi^{i\prime}_{t+1}(y) + C^i(y) + 2\pi^i_{t+1}(y) - 2p^i\right].\end{aligned}$$

Using the fisherman's species-specific optimality condition $C^i(Y^i(\tau)) = p^i - \tau$,

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= y^3\left[y\pi^{i\prime}_{t+1}(y) + p^i - \tau + 2\pi^i_{t+1}(y) - 2p^i\right] \\ &= \underbrace{y^3}_{>0}\underbrace{\left[y\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) - p^i - \tau\right]}_{\text{sign of } \hat{w}^{i\prime\prime}}.\end{aligned}$$

Since $y^3 > 0$, $\hat{w}^{i\prime\prime}(z) \leq 0$ if and only if the bracket is $\leq 0$, which gives:

**Condition for global concavity.** For every $i$ and every $\tau \in [\tau^{1*}, \tau^{N*}]$, writing $y = Y^i(\tau)$,

$$y\,\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) \;\leq\; p^i + \tau, \tag{46}$$

with strict inequality for at least one $i$.

Because $W$ is separable, imposing (46) for every $i$ makes $W''(\tau) < 0$.

## Appendix D.4: Interiority

The endpoint signs $g(\tau^{1*}) < 0 < g(\tau^{N*})$ are strict, so $\tau^+ \in (\tau^{1*}, \tau^{N*}) \subset (0, \tau^F_{\max})$. The optimum is therefore never at $\min_i \tau^{i*}$ or $\max_i \tau^{i*}$ unless these coincide.

## Appendix E: Derivation of the optimal quota basket

### E.1: Fisherman's problem

Each period, the fisherman maximizes profits over the harvest of $N$ stocks subject to the shared quota:

$$\begin{aligned} \max_{\{h_t^i\}} \left\{ \pi_t = \sum_i \Big( p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s^i)\, ds^i \Big) \right\} \\ \text{subject to} \quad \omega_t = \sum_i h_t^i = \sum_i X_t^i - \sum_i y_t^i \end{aligned} \tag{47}$$

The Lagrangian is

$$\mathcal{L} = \sum_i \Big( p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s^i)\, ds^i \Big) + \lambda_t \Big( \omega_t - \sum_i h_t^i \Big),$$

with FOC for each stock

$$\begin{aligned} p^i - C^i(y_t^i) &= \lambda_t \\ y_t^i &= (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t) \end{aligned} \tag{48}$$

The realized stocks $X_t^i = Z_t^i G^i(y_{t-1}^i)$ depend on last period's escapement and this period's shock, not on the current quota:

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\ \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \lambda_t &= \lambda(\mathbf{X}_t, \omega_t) \end{aligned} \tag{49}$$

Differentiating (49) with respect to $\omega_t$, holding the stocks $\mathbf{X}_t$ fixed (they are predetermined when the fisherman harvests),

$$\begin{aligned} \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \frac{\partial}{\partial \omega_t} \Big( \sum_i Y^i(\lambda_t) \Big) &= \frac{\partial}{\partial \omega_t} \Big( \sum_i X_t^i - \omega_t \Big) \\ \Big( \sum_i Y_\lambda^i \Big) \lambda_{\omega_t} &= -1 \\ \lambda_{\omega_t} &= \frac{-1}{\sum_i Y_\lambda^i} < 0 \end{aligned} \tag{50}$$

If the cost function is $C^i = \theta^i/s^i$, $Y^i_\lambda = \theta^i/(p^i-\lambda)^2 = (y^i)^2/\theta^i > 0$, because $p^i - \lambda = C^i(y^i) = \theta^i/y^i$ and $y^i, \theta^i > 0$. Consequently, a larger quota lowers the common shadow value. A change in the harvest limit causes the following change in each species' escapement:

$$\begin{aligned} y^i_t &= Y^i(\lambda_t) \\ \frac{\partial y^i_t}{\partial \omega_t} &= Y^i_\lambda \, \lambda_{\omega_t} \\ &= Y^i_\lambda \ \frac{-1}{\sum_j Y^j_\lambda} \qquad \text{Replace (50)} \\ &= -\frac{(y^i_t)^2/\theta^i}{\sum_j (y^j_t)^2/\theta^j} \\ &= -\rho^i_t \qquad \text{Rename} \end{aligned} \tag{51}$$

By construction, $\sum_i \rho^i_t = 1$; the regularity condition ensures that every species enters the sum. The weights $\rho^i_t$ record how the fisherman spreads one extra unit of the quota basket across species.

### E.2: Choosing the quota is choosing the shared shadow value

Total profits per period are:

$$\Pi(\mathbf{X}_t, \omega_t) := \sum_i \int_{y^i_t}^{X^i_t} \left(p^i - C^i(s)\right) ds$$

with $y^i_t = Y^i(\lambda_t)$. Differentiating with respect to $\omega_t$, and using the fisherman's FOC $p^i - C^i(y^i_t) = \lambda_t$ and $\sum_i \partial y^i_t/\partial \omega_t = -1$ from (49),

$$\begin{aligned} \frac{\partial \Pi}{\partial \omega_t} &= -\sum_i \left(p^i - C^i(y^i_t)\right) \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\sum_i \lambda_t \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\lambda_t \sum_i \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\lambda_t \sum_i (-\rho^i_t) \\ &= \lambda_t \end{aligned} \tag{52}$$

The marginal social value of an extra unit of aggregate quota equals the fisherman's shadow value $\lambda_t$. This is the duality between the two derivations: $\lambda_t$ is the price that decentralizes the quota.

### E.3: Social planner's problem

Take the predetermined escapement vector $\vec{y}_{t-1}$ as the state. The planner sets the quota $\omega_t$ before the shocks, so $\omega_t$ is deterministic while $\lambda_t$ and $y_t^i$ are random through $\mathbf{X}_t$:

$$V(\vec{y}_{t-1}) = \max_{\omega_t \geq 0} \mathbb{E}\Big[\Pi(\mathbf{X}_t, \omega_t) + \beta V(\vec{y}_t)\Big], \qquad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad y_t^i = Y^i\big(\lambda(\mathbf{X}_t, \omega_t)\big) \tag{53}$$

Only $y_t^i$ depends on $\omega_t$ (the stocks $\mathbf{X}_t$ are predetermined by $\vec{y}_{t-1}$). Using $\partial\Pi/\partial\omega_t = \lambda_t$ from (52) and $\partial y_t^i/\partial\omega_t = -\rho_t^i$ from (51), I derive the FOC with respect to the quota basket:

$$\begin{aligned} \frac{\partial V(\cdot)}{\partial \omega_t} &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta[V_{y^1} y_\lambda^1 \lambda_\omega + V_{y^2} y_\lambda^2 \lambda_\omega + \cdots + V_{y^N} y_\lambda^N \lambda_\omega]\Big] = 0 \\ &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta \sum_j V_{y^j}(\vec{y}_t)\, \frac{\partial y_t^j}{\partial \omega_t}\Big] = 0 \\ &= \mathbb{E}\Big[\lambda_t - \beta \sum_j \rho_t^j\, V_{y^j}(\vec{y}_t)\Big] = 0 \end{aligned} \tag{54}$$

The optimal quota basket equates, in expectation, the current shadow value of the quota with the $\rho$-weighted continuation value of the escapement.

The state $y_{t-1}^k$ enters (53) through $X_t^k = Z_t^k G^k(y_{t-1}^k)$. Differentiating (49) with respect to the stock, holding $\omega_t$ fixed,

$$\Big(\sum_j Y_\lambda^j\Big)\frac{\partial \lambda_t}{\partial X_t^k} = 1, \qquad \frac{\partial y_t^j}{\partial X_t^k} = Y_\lambda^j \frac{\partial \lambda_t}{\partial X_t^k} = \frac{Y_\lambda^j}{\sum_m Y_\lambda^m} = \rho_t^j, \qquad \sum_j \frac{\partial y_t^j}{\partial X_t^k} = 1 \tag{55}$$

Applying the envelope theorem (the indirect effect through the optimal $\omega_t$ is zero by (54)) and the chain rule,

$$
\begin{aligned}
V_{y^k_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(\frac{\partial \Pi}{\partial X^k_t} + \beta \frac{\partial V(\vec{y}_t)}{\partial X^k_t}\Big) \underbrace{Z^k_t G^{k\prime}(y^k_{t-1})}_{\partial X^k_t / \partial y^k_{t-1}}\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\frac{\partial X^i_t}{\partial X^k_t} - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial \lambda_t}\frac{\partial \lambda_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\delta_{ik} - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t \sum_j \rho^j_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big]
\end{aligned}
\tag{56}
$$

using $\partial X^i_t / \partial X^k_t = \delta_{ik}$ for the stocks (only stock $k$ responds to its own $y^k_{t-1}$) and $\partial y^j_t / \partial X^k_t = \rho^j_t$ from (55) for the quota-coupled escapements (all respond, through $\lambda_t$). The bracket is the value of one extra fish in stock $k$, minus the shadow value $\lambda_t$ of the escapement it displaces, plus its continuation value. The escapements are all tied together through the quota basket.

Equations (54)–(56) are the direct-$\omega$ analog of Appendix B.3.

### E.4: Recovering $\lambda^+$

With $Z^i_t \equiv 1$, the expectation drops and (54) holds pointwise, $\lambda_t = \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)$. Substituting this into (56) cancels the last two bracketed terms, leaving

$$
\begin{aligned}
V_{y^i_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}_{Z_t}\Big[\Big(p^i - C^i(X^i_t) - \lambda_t + \beta \sum_j \rho^j_t\, V_{y^j}(\vec{y}_t)\Big) Z^i_t G^{i\prime}(y^i_{t-1})\Big] \\
&= \big(p^i - C^i(X^i_t) - \lambda_t + \lambda_t\big) G^{i\prime}(y^i_{t-1}) \\
&= \big(p^i - C^i(G^i(y^i_{t-1}))\big) G^{i\prime}(y^i_{t-1}) \\
&= \frac{1}{\beta}\pi^i_{t+1}(y^i_{t-1}) \quad \text{using the definition } \pi^i_{t+1} = \beta\big(p^i - C^i(G^i)\big)G^{i\prime} \text{ with } Z \equiv 1
\end{aligned}
\tag{57}
$$

Substituting back into the FOC and using Remark 1.2,

$$
\begin{aligned}
\lambda_t &= \beta \sum_i \rho_t^i V_{y^i}(\vec{y}_t) \\
\lambda_t &= \sum_i \rho_t^i \, \pi_{t+1}^i(y_t^i) \\
\lambda^+ &= \frac{\sum_i \pi_{t+1}^i(y_t^i)\,(y_t^i)^2/\theta^i}{\sum_j (y_t^j)^2/\theta^j}
\end{aligned}
\tag{58}
$$

which is exactly the planner's FOC for the tax, (14). Like the optimal tax basket, the optimal shadow value is fixed over time, $\lambda_t = \lambda^+$, with $y_t^i = Y^i(\lambda^+)$, if the regularity condition holds.

### E.5: The quota basket is second-best

Under uncertainty, the collapse in Appendix E.4 fails because the shadow value $\lambda_t^F$ and the reallocation weights $\rho_t^i$ are both functions of the realized shocks. The planner, by contrast, commits to a deterministic $\omega_t$. The FOC residual is

$$
R_t := \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t),
$$

so that (54) reads $\mathbb{E}[R_t] = 0$. Substituting into (56) and separating,

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - R_t\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\big[(p^k - C^k(X_t^k))Z_t^k\big] G^{k\prime}(y_{t-1}^k) - \mathbb{E}\big[R_t Z_t^k\big] G^{k\prime}(y_{t-1}^k)
\end{aligned}
$$

With $C^k(s) = \theta^k/s$ and $X_t^k = Z_t^k G^k(y_{t-1}^k)$, the shock cancels in the first expectation:

$$
\mathbb{E}\Big[\Big(p^k - \frac{\theta^k}{Z_t^k G^k(y_{t-1}^k)}\Big) Z_t^k\Big] = p^k - \frac{\theta^k}{G^k(y_{t-1}^k)} = p^k - C^k\big(G^k(y_{t-1}^k)\big)
$$

The second expectation, however, is a covariance, because $\mathbb{E}[R_t] = 0$ and $\mathbb{E}[Z_t^k] = 1$:

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \frac{1}{\beta}\pi_{t+1}^k(y_{t-1}^k) - \operatorname{Cov}\big(R_t,\, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k) \\
\beta V_{y^k}(\vec{y}_{t-1}) &= \pi_{t+1}^k(y_{t-1}^k) - \beta \operatorname{Cov}\big(R_t,\, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k)
\end{aligned}
\tag{59}
$$

In a deterministic setting, $\operatorname{Cov}(R_t, Z_t^k) = 0$ and (59) reduces to $\beta V_{y^k} = \pi_{t+1}^k$, which is Appendix E.4. Under uncertainty, the covariance is nonzero, because $R_t$ moves with the

shocks through both $\lambda_t^F$ and $\rho_t^j$. The planner's single scalar $\omega_t$ can set the mean of $R_t$ to zero, but it cannot remove its correlation with the shock vector. Given the deterministic quota,

$$\sum_i Y^i(\lambda_t^F) = \sum_i Z_t^i G^i(y_{t-1}^i) - \omega_t,$$

so $\lambda_t^F$ inherits the randomness of the aggregate realized stock. The quota choice

$$\omega_t = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\lambda^+) \tag{60}$$

uses only $\mathbb{E}[Z_t^i] = 1$ and the predetermined $y_{t-1}^i$. It centers the distribution of $\lambda_t^F$ near $\lambda^+$ and the escapements near their targets $\bar{y}^i = Y^i(\lambda^+)$; it is the quota the planner can implement before the shocks are realized.

## Appendix F: the bounds of the welfare loss

**Lemma 2** (The welfare loss is bounded)**.** *$w^i(y) = \phi^i(y) + \beta a^i(y)$ is the dynamic per-species payoff of (41), with $\phi^i(y) = -p^i y + \theta^i \ln y$ and $a^i(y) = p^i G^i(y) - \theta^i \ln G^i(y)$. By (23), $w^{i\prime}(y) = \pi^i_{t+1}(y) - (p^i - C^i(y))$, so $w^{i\prime}(\bar{y}^i) = 0$, and $w^{i\prime\prime}(y) < 0$ on the fishable range $[\underline{y}^i, \hat{y}^i]$, where the bounds are the smallest and largest escapements the quota can induce (see Lemma 3), so that $0 < \underline{y}^i \leq \bar{y}^i \leq \hat{y}^i < \infty$. Then, for every $y \in [\underline{y}^i, \hat{y}^i]$:*

*(a) The loss relative to the basket optimum is non-negative:*

$$0 \;\leq\; w^i(\bar{y}^i) - w^i(y). \tag{61}$$

*(b) The loss is at most a constant times the squared deviation:*

$$w^i(\bar{y}^i) - w^i(y) \;\leq\; \tilde{B}^i\,(y - \bar{y}^i)^2, \qquad \tilde{B}^i = \tfrac{1}{2}\max_{[\underline{y}^i, \hat{y}^i]}\left(-w^{i\prime\prime}\right) < \infty. \tag{62}$$

The global maximizer of the concave $\bar{y}^i$ verifies (a). For part (b), I know:

$$w^i(b) - w^i(y) = \int_y^b w^{i\prime}(s)ds$$

renaming $u = w^{i\prime}(s)$ and $v = s - b$, then by integration by parts:

$$\begin{aligned}
\int_y^b w^{i\prime}(s)ds &= \Big[w^{i\prime}(s)(s-b)\Big]\Big|_b^y - \int_y^b (s-b)w^{i\prime\prime}(s)ds \\
&= -w^{i\prime}(y)(y-b) - \int_y^b (s-b)w^{i\prime\prime}(s)ds \\
&= w^{i\prime}(y)(b-y) - \int_y^b (s-b)w^{i\prime\prime}(s)ds
\end{aligned}$$

if $b = \bar{y}^i$, then $w^{i\prime} = 0$, then

$$w^i(\bar{y}^i) - w^i(y) = \int_y^{\bar{y}^i} \left(-w^{i\prime\prime}(s)\right)(s-y)\,ds,$$

The integrand is non-negative because $w^{i\prime\prime} < 0$. I control the curvature of $w^{i\prime\prime}$ with the maximum possible value in the range:

$$M^i := \max_{[\underline{y}^i, \hat{y}^i]}(-w^{i\prime\prime})$$

Then, for every $s$, $-w^{i\prime\prime}(s) \leq M^i$. Since $(s-y) \geq 0$, then

$$-w^{i\prime\prime}(s)(s-y) \leq M^i(s-y)$$

Integrating both sides

$$\begin{aligned}
-w^{i\prime\prime}(s)(s-y) &\leq M^i(s-y) \\
\int_y^{\bar{y}^i} -w^{i\prime\prime}(s)(s-y)ds &\leq M^i \int_y^{\bar{y}^i} (s-y)ds \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \int_y^{\bar{y}^i} (s-y)ds \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \int_y^{\bar{y}^i - y} u du \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \frac{u^2}{2}\Big|_{\bar{y}^i - y}^0 \\
w^i(\bar{y}^i) - w^i(y) &\leq M^i \frac{(\bar{y}^i - y)^2}{2} \\
0 \leq w^i(\bar{y}^i) - w^i(y) &\leq \tilde{B}^i (y - \bar{y}^i)^2
\end{aligned}$$

The maximum is finite because $w^{i\prime\prime}$ is continuous and no species can be driven to extinction due to the regularity condition.

The next lemma shows the shadow-value deviation is also bounded:

**Lemma 3** (Bounds for the fisherman's response)**.** *Let $Y^i(\lambda) = \theta^i/(p^i - \lambda)$ be the fisherman's escapement response under a quota. Suppose the realized shadow value is confined to a bounded interval $\lambda_t^F \in [\lambda_L, \lambda_u]$ with $\lambda_u < \min_i p^i$. Then there exist finite positive constants*

$$\underline{m} := \min_i \frac{\theta^i}{(p^i - \lambda_L)^2} > 0, \qquad \overline{m}^i := \frac{\theta^i}{(p^i - \lambda_u)^2} < \infty, \tag{63}$$

*such that, for any two shadow values $\lambda, \lambda' \in [\lambda_L, \lambda_u]$,*

$$\underline{m}\,|\lambda - \lambda'| \;\leq\; \left|Y^i(\lambda) - Y^i(\lambda')\right| \;\leq\; \overline{m}^i\,|\lambda - \lambda'|. \tag{64}$$

*In particular, taking $\lambda = \lambda_t^F$, $\lambda' = \lambda^+$ and $y_t^i = Y^i(\lambda_t^F)$, $\bar{y}^i = Y^i(\lambda^+)$,*

$$\underline{m}\,|\lambda_t^F - \lambda^+| \;\leq\; |y_t^i - \bar{y}^i| \;\leq\; \overline{m}^i\,|\lambda_t^F - \lambda^+|. \tag{65}$$

The response function has derivative

$$Y^{i\prime}(\lambda) = \frac{d}{d\lambda}\frac{\theta^i}{p^i - \lambda} = \frac{\theta^i}{(p^i - \lambda)^2} > 0,$$

so $Y^i$ is strictly increasing. Differentiating once more, $Y^{i\prime\prime}(\lambda) = 2\theta^i/(p^i - \lambda)^3 > 0$ on $\lambda < p^i$, so $Y^{i\prime}$ is itself increasing in $\lambda$. Then, the response is flattest at the lower end $\lambda_L$ and steepest at the upper end $\lambda_u$. Hence on $[\lambda_L, \lambda_u]$,

$$\frac{\theta^i}{(p^i - \lambda_L)^2} \;\leq\; Y^{i\prime}(\lambda) \;\leq\; \frac{\theta^i}{(p^i - \lambda_u)^2} = \overline{m}^i$$

Taking the minimum over species gives the floor $\underline{m}$ in (63). Both constants are finite and strictly positive because $\lambda_u < \min_i p^i$ avoids the denominator becoming 0.

By the mean value theorem, for any $\lambda, \lambda'$ there is a point $\zeta^i$ between them with $Y^i(\lambda) - Y^i(\lambda') = Y^{i\prime}(\zeta^i)(\lambda - \lambda')$. Since $\zeta^i \in [\lambda_L, \lambda_u]$, its slope lies between the two bounds, which gives (64).